Robust Sublinear Convergence Rates for Iterative Bregman Projections

Anonymous Author(s)
Affiliation
Address
`email`

Abstract

Entropic regularization provides a simple way to approximate linear programs whose constraints split into two or more tractable blocks. The resulting objectives are amenable to cyclic Kullback–Leibler (KL) Bregman projections, with Sinkhorn-type algorithms for optimal transport, matrix scaling, and barycenters as canonical examples. This paper gives a general blueprint for proving $O(1/k)$ dual convergence rate with a constant that scales only *linearly* in $1/\gamma$, where γ is the entropic regularization parameter. We call such rates "robust", because this mild dependence on γ underpins favorable complexity bounds for approximating the unregularized problem via alternating KL projections. The blueprint reduces the proof to a uniform primal bound and a dual bound for a quotient norm induced by the constraint split. To make these inputs usable, we propose two helper results, which rely on the non-expansiveness of the dual iterations in this quotient dual norm. Instantiating this blueprint for graph-structured transport yields a new flow-Sinkhorn algorithm for the Wasserstein-1 distance on graphs. It achieves ε-additive accuracy on the transshipment cost in $O(p\,\mathrm{diameter}^3/\varepsilon^4)$ arithmetic operations (up to logarithmic factors), where p is the number of edges. We also provide a machine-checked Lean formalization of the paper-facing theorem statements, including a Comparator-ready challenge/solution layer for independent statement checking.

1 Introduction

Iterative Bregman projections power many scalable ML solvers: they turn entropy-regularized constrained problems into alternating normalization steps implemented with sparse linear algebra and GPU-friendly tensor operations. They underlie Sinkhorn-type methods for optimal transport [Cuturi, 2013] and their generalizations, such as barycenters [Benamou et al., 2015] and unbalanced-OT [Chizat et al., 2018]. This paper provides a blueprint for proving sublinear convergence with constants that remain "robust" at small regularization. The Wasserstein-1 distance on graphs is one concrete instance of this blueprint: the graph structure changes the split and the dual quotient norm with respect to the vanilla Sinkhorn analysis, but the proof still follows the same route.

Entropic regularization of structured linear programs. We consider a feasible linear program $\min_{Ax=b,\, x\geq 0}\langle x, C\rangle$ whose constraints $Ax = b$ naturally split into blocks. We focus on the two-block case for clarity. Throughout, let $d \in \mathbb{N}$ be the dimension of the problem and write $\mathbb{R}^d_+$ for the positive orthant over which the optimization is carried out. Let $A = (A_1; A_2)$, with $A_1 \in \mathbb{R}^{m_1\times d}$, $A_2 \in \mathbb{R}^{m_2\times d}$ and $m = m_1 + m_2$, and split $b \in \mathbb{R}^m$ as $b = (b_1; b_2)$ with $b_1 \in \mathbb{R}^{m_1}$ and $b_2 \in \mathbb{R}^{m_2}$. We define the two affine constraint sets

$$\mathcal{C}_1 \coloneqq \{x \in \mathbb{R}^d_{++} : A_1 x = b_1\}, \qquad \mathcal{C}_2 \coloneqq \{x \in \mathbb{R}^d_{++} : A_2 x = b_2\}.$$

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

We assume $\mathcal{C}_1 \cap \mathcal{C}_2 \neq \varnothing$, and in particular $b \in \mathrm{range}(A)$. This linear program can thus be re-written as

$$\min_{x\in\mathbb{R}^d_+} \{\langle C, x\rangle \quad \text{s.t.} \quad A_1 x = b_1, \;\; A_2 x = b_2\} \;=\; \min_{x\in\mathcal{C}_1\cap\mathcal{C}_2} \langle C, x\rangle \tag{$\mathcal{P}_0$}$$

for some cost $C \in \mathbb{R}^d$. Fix a reference vector $z \in \mathbb{R}^d_{++}$. For a temperature $\gamma > 0$, we consider the following entropically regularized problem, which aims to approximate the original feasible linear program using fast iterative schemes

$$\min_{x\in\mathcal{C}_1\cap\mathcal{C}_2} \langle C, x\rangle + \gamma\,\mathrm{KL}(x|z), \quad \text{where} \quad \mathrm{KL}(x|z) \coloneqq \sum_{i=1}^d \Big(x_i \log\frac{x_i}{z_i} - x_i + z_i \Big). \tag{$\mathcal{P}_\gamma$}$$

It is common to use a constant reference $z = \alpha\mathbb{1}_d$ for some scale parameter $\alpha \in \mathbb{R}_+$. Since we do not restrict our attention to probability vectors x (as in optimal transport), the choice of α is meaningful and should reflect prior knowledge about the expected total mass of the solution. The regularised objective admits a simple *cyclic Bregman projection* scheme detailed in Section 2: project x alternately onto $\mathcal{C}_1$ and $\mathcal{C}_2$ in the Kullback–Leibler divergence. This method is attractive when both projections are explicit; the classical Sinkhorn algorithm for optimal transport is the canonical example.

This paper establishes sharp sub-linear $O(1/(\gamma k))$ convergence bounds for the *dual objective* of the KL-projection scheme and demonstrates their impact through a new algorithm for the Wasserstein–1 distance on graphs.

Iterative Bregman projection. The idea of alternating Bregman projections originates from the work of Bregman [Bregman, 1967] and of Csiszár–Tusnády [Csiszár and Tusnády, 1984]. When the distance is squared Euclidean, this procedure reduces to von Neumann's alternating projections, whose linear convergence under an angle condition was already established by Friedrichs [Friedrichs, 1938]. More generally, asymptotic linear rates for Bregman projections were shown in [Censor and Rezač, 2015], again assuming a qualification property that, in Euclidean settings, corresponds to such an angle condition. This assumption fails for the KL divergence unless all coordinates of the iterates remain bounded away from zero, something that, in many applications such as entropically regularised optimal transport, would require a bound of order $1 - e^{-\|C\|_\infty/\gamma}$, which becomes prohibitively small when γ is small. These earlier convergence results focus on bounding the distance between primal iterates and the solution, typically in ℓ^1 norm for the KL divergence. Our interest instead lies in controlling the *dual* value, which provides a weaker guarantee but leads to sharper constants, avoiding the exponential blow-up in $1/\gamma$ and making the analysis relevant in high-dimensional machine learning contexts.

Entropic regularisation and Sinkhorn-type methods. The link between entropy and transport can be traced back to Schrödinger's 1931 formulation of the Brownian bridge problem. In the modern computational setting, Cuturi [Cuturi, 2013] popularised entropic regularisation for large-scale optimal transport, showing that the resulting smoothed problems bypass the curse of dimensionality in empirical settings. The associated iterative scaling scheme was introduced independently multiple times: it appears as early as Yule's work in 1912 [Yule, 1912] and was later introduced and analyzed by Sinkhorn [Sinkhorn, 1964], as well as by Deming–Stephan [Deming and Stephan, 1940]. As reviewed in [Chizat et al., 2025], convergence analyses for these iterations generally fall into two categories. The first comprises *linear rates* (i.e., exponential with the iteration index k) results with constants that degrade exponentially badly when γ is small, which we call non-robust rates following [Chizat et al., 2025]. One approach uses Hilbert's projective metric to show contraction [Franklin and Lorenz, 1989, Borwein et al., 1994], with rates of the order $[1 - e^{-\|C\|_\infty/\gamma}]^k$; see [Chen et al., 2016, Deligiannidis et al., 2024] for continuous-domain extensions and [Eckstein, 2025] for non-compact domains. Equivalent dependencies arise from convex optimisation proofs [Marino and Gerolin, 2020, Carlier, 2022], which extend to multi-marginal settings [Greco et al., 2023, Conforti et al., 2023]. For arbitrary, possibly adversarial, cost, this dependence on $\|C\|_\infty/\gamma$ appears tight. The second category covers polynomial-rate bounds (typically $1/k$) with constants that remain stable as γ changes (what we call "robust" rates). Early complexity results of this type go back to [Kalantari et al., 2008] and were sharpened in later works [Altschuler et al., 2017, Chakrabarty and Khanna, 2021, Dvurechensky et al., 2018, Chizat et al., 2020]. One obtains rate on the dual objective of the order $\|C\|^2_\infty/(\gamma k)$. This approach can be reframed as mirror descent in a tailored geometry [Léger, 2021, Aubin-Frankowski et al., 2022]. In continuous cases (or in discrete cases if one accepts dependency on the number of Dirac masses), it is, however, possible to obtain the best of both worlds, and [Chizat et al., 2025] shows that when the marginals are bounded from below that the linear rate is of the order $[1-\kappa\gamma^2/C_{\max}^2]^k$ for some constant κ. Our contributions focus on sublinear rates, and aim at understanding the general structure that enables robust rates, to apply this analysis to a larger class of linear programs.

Wasserstein-1 on graphs. The Wasserstein-1 distance has a particular structure that makes it both more robust and more tractable than W_2: it is less sensitive to outliers, and it admits a flow formulation that is often faster to compute. It has been applied in computer vision [Rubner et al., 1998, Grauman and Darrell, 2004], machine learning [Kusner et al., 2015], graphics [Solomon et al., 2014], community detection [Sia et al., 2019], and biology [Sandhu et al., 2015]. When the ground cost is the shortest-path distance on a graph, W_1 can be formulated as a minimum-cost flow problem with a fixed divergence constraint [Beckmann, 1952], see for instance [Carlier and Santambrogio, 2012] for applications of this framework to PDEs on continuous domains. Classical solvers such as the network simplex can be used, and Orlin's strongly polynomial algorithm achieves $O(pn\log n)$ time on a graph with n vertices and p edges [Ahuja et al., 1993]. The current best exact min-cost flow algorithm runs in $O(p^{1+o(1)})$ time with high probability [Chen et al., 2025]. For planar graphs with polynomially bounded integer data, a nearly linear-time exact algorithm is known [Dong et al., 2025], and for tree metrics (generalizing the 1-D case), W_1 can be computed in linear time $O(n)$ [Evans and Matsen, 2012]. For approximate solvers, which are the focus of this paper, interior-point methods combined with fast Laplacian solvers yield additive-ε approximations for min-cost generalized flows in $O(p^{3/2}\log(1/\varepsilon))$ time [Daitch and Spielman, 2008]. In this work, we propose a simpler alternative based on entropic regularization, with complexity $O(p\,\mathrm{diameter}(E)^3/\varepsilon^4)$ (up to logarithmic factors). While this has a worse dependence on ε, it is easy to implement (including on GPUs) and scales linearly with the number of edges p.

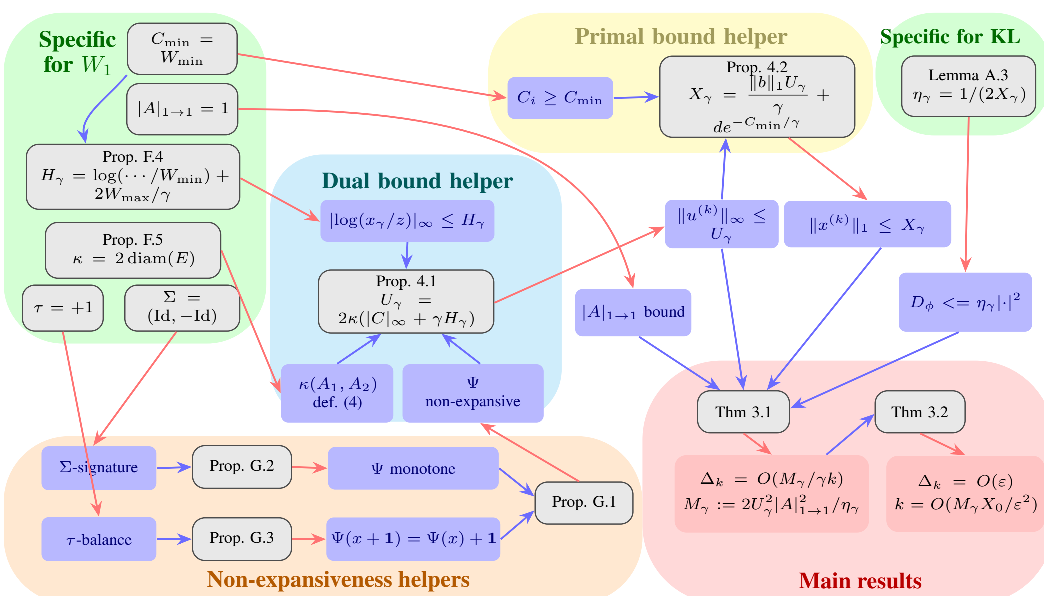

Figure 1: Logical map of the convergence blueprint and of its KL/flow-Sinkhorn instantiations. Blue arrows indicate hypotheses used by a theorem, while red arrows indicate the result statement of that theorem.

Contributions. The core contribution of this paper is a general blueprint for analyzing the convergence speed of iterative Bregman projection methods summarized in Figure 1, which presents the architecture of the paper and how the abstract blueprint is instantiated for the graph-W_1 flow-Sinkhorn algorithm. The main convergence results, shown in red, are Theorem 3.1 and Theorem 3.2. They rely on primal and dual bounds handled by the helper results in yellow and cyan, including Proposition 4.2 and Proposition 4.1, and crucially on non-expansiveness of the dual update map Ψ, treated by the light-orange helper results in Appendix G. The graph-W_1 instantiation, shown in green, yields the second core contribution detailed in Section 5: the flow-Sinkhorn algorithm, whose sparse updates scale linearly with the number of edges and lead to an explicit additive-accuracy complexity guarantee, together with numerical diagnostics on synthetic and genomic-inspired sparse graphs, including GPU line-scaling experiments, that illustrate the regularization/runtime tradeoff in the unregularized-accuracy regime. A third contribution is the Lean formalization described in Appendix H: every theorem, proposition, lemma, and corollary in the manuscript is linked to a compiled Lean endpoint through a stable paper-facing map, and the repository includes a Comparator-ready challenge/solution layer for independent statement checking. The implementation, benchmark scripts, supplementary material, and Lean development are included in the submitted artifacts. Appendices G and C record auxiliary non-expansiveness results and general Bregman extensions.

2 Iterative KL Projections

We begin by casting any linear program with two affine constraints into an *entropically regularised* form. The resulting objective is minimised by the classical *cyclic KL–projection* (a.k.a. iterative Bregman projection) algorithm, whose dual convergence is analysed in Section 3—culminating in the $O(1/k)$ rate of Theorem 3.1. Throughout the section, we keep the presentation self-contained and focus on two blocks for clarity.

Entropic regularisation and cyclic KL projections. Problems $(\mathcal{P}_\gamma)$ can be conveniently re-written as a Bregman projection problem of the tilted reference z^C (also called Gibbs kernel for OT problems)

$$\min_{x\in\mathcal{C}_1\cap\mathcal{C}_2} \mathrm{KL}(x|z^C), \qquad z_i^C := z_i e^{-C_i/\gamma}. \tag{1}$$

The vector z^C is the Gibbs kernel associated with C, γ and z. Formulation (1) reveals the structure exploited in the sequel: alternating KL projections onto $\mathcal{C}_1$ and $\mathcal{C}_2$ provides a scalable solver whose convergence is quantified in Section 3.

Since the constraints are affine, problem (1) can be solved by iterative Bregman projection using the KL Bregman divergence. Starting from $x^{(0)} = z^C$, it performs the two-step sweep

$$x^{(k+\frac{1}{2})} := \operatorname*{arg\,min}_{x\in\mathcal{C}_1} \mathrm{KL}(x|x^{(k)}) \quad (P_1), \qquad x^{(k+1)} := \operatorname*{arg\,min}_{x\in\mathcal{C}_2} \mathrm{KL}(x|x^{(k+\frac{1}{2})}) \quad (P_2). \tag{2}$$

When each projection has a closed form, or can be computed with only a few inexpensive inner iterations, (P_1)–(P_2) provide an efficient outer loop. For instance, Appendix E revisits the classical optimal-transport splitting that yields Sinkhorn's method, while Section 5 introduces a new splitting for the Wasserstein-1 flow formulation on graphs.

Dual problem. We now express the entropic programme $(\mathcal{P}_\gamma)$ in the dual variables by introducing the dual functional F_γ, which is the quantity used throughout the paper to measure convergence speed.

Proposition 2.1 (Dual of $(\mathcal{P}_\gamma)$)**.** *Let $z \in \mathbb{R}^d_{++}$ be fixed and set $Z := \sum_{i=1}^d z_i$. Assume the following finite primal–dual certificate holds for some $u^\star$: weak duality between the displayed primal objective and the dual objective below, feasibility of $x(u^\star)$, zero primal–dual gap at this pair, uniqueness of the primal optimizer by objective value, and the gradient identity $\nabla F_\gamma(u^\star) = b - Ax(u^\star)$. Then the certified primal and dual values satisfy*

$$\min(P_\gamma) = \max_{u\in\mathbb{R}^m} F_\gamma(u) := \langle b, u\rangle + \gamma Z - \gamma \sum_{i=1}^d z_i \exp\Big(\tfrac{(A^\top u)_i - C_i}{\gamma}\Big), \tag{$\mathcal{D}_\gamma$}$$

where $\min(P_\gamma)$ denotes the value of $(\mathcal{P}_\gamma)$. Moreover, any maximiser $u^\star$ and the unique primal minimiser $x^\star$ are linked by $x^\star = x(u^\star)$, where $x(u)_i := z_i \exp(((A^\top u)_i - C_i)/\gamma) = z_i^C \exp((A^\top u)_i/\gamma)$, $i = 1,\dots,d$, and $u^\star$ is characterised by the stationarity condition $\nabla F_\gamma(u^\star) = 0 \Longleftrightarrow A\,x(u^\star) = b$.

Proposition 2.1 allows us to translate the cyclic projections (P_1)–(P_2) into a block-coordinate ascent on F_γ as we detail next. Note that while dual maximizers might not be unique, they are unique up to $\ker(A^\top)$, which is important to take into account in the analysis of the algorithm (for classical OT, this corresponds to translation of the dual potential).

Block–coordinate ascent in the dual. With the split $u = (u_1, u_2) \in \mathbb{R}^{m_1} \times \mathbb{R}^{m_2}$ we write $F_\gamma(u) = F_\gamma(u_1, u_2)$. The two primal KL projections (P_1)–(P_2) are exactly one sweep of block maximisation on F_γ: $u_1^{(k+1)} = \Psi_1(u_2^{(k)})$ and $u_2^{(k+1)} = \Psi_2(u_1^{(k+1)})$, where $\Psi_1(u_2) := \arg\max_{u_1 \in \mathbb{R}^{m_1}} F_\gamma(u_1, u_2)$ and $\Psi_2(u_1) := \arg\max_{u_2 \in \mathbb{R}^{m_2}} F_\gamma(u_1, u_2)$, and the corresponding primal variable is obtained from Proposition 2.1: $x^{(k)} = x(u_1^{(k)}, u_2^{(k)})$. We define the full sweep map on the u_1 variable as $\Psi := \Psi_1 \circ \Psi_2$ so that $u_1^{(k)} = \Psi^k(u^{(0)})$. For several important splittings, for instance, the classical OT split in Appendix E and the flow split in Section 5, these maps have closed-form solutions, which makes this algorithm practical.

3 Dual convergence

The cyclic projections of Section 2 are equivalent to block–coordinate ascent on the dual objective F_γ introduced in Proposition 2.1. This section proves the two main red-box results summarized in Figure 1: Theorem 3.1, which gives a robust $O(1/(\gamma k))$ dual convergence rate, and Theorem 3.2, which converts this rate into an additive-accuracy guarantee for the unregularized linear program. The point of the blueprint is to make these robust rates modular: once primal and dual boundedness are available, the convergence proof is automatic, and Section 4 explains how the helper results obtain these bounds from non-expansiveness of the sweep map in the relevant quotient dual norm.

Definition 3.1 (block-quotient seminorms)**.** *For $u = (u_1, u_2)$ dual variables, we define $\|u_1\|_{V_1} := \inf_{h \in \ker(A^\top)} \|u_1 + h_1\|_\infty$ and $\|u_2\|_{V_2} := \inf_{h \in \ker(A^\top)} \|u_2 + h_2\|_\infty$. The associated block-quotient dual semi-norm, which depends on the split of constraints (A_1, A_2), is $\|u\|_V := \max\{\|u_1\|_{V_1}, \|u_2\|_{V_2}\}$. Note that in the special case of classical OT (see Appendix E), this corresponds to the so-called variation semi-norm defined in* (26)*.*

The crucial hypotheses for Theorem 3.1 are bounded primal iterates in ℓ^1 norm and bounded dual iterates in the block-quotient semi-norm $\|\cdot\|_V$. The blueprint is designed to establish these hypotheses through two helpers: a dual bound helper based on non-expansiveness of the sweep map in the quotient norm, and a primal bound helper built on top of that dual control. This is where the split (A_1, A_2) enters the analysis, since it determines the quotient geometry used to measure dual radius.

Theorem 3.1 (Sub–linear dual rate)**.** *Let $(\Delta_k)_{k \geq 0}$ be a nonnegative dual-gap sequence and let $(r_k)_{k \geq 0}$ be a residual sequence. Assume $\gamma > 0$, $X_\gamma > 0$, $U_\gamma > 0$ and $\|A\|_{1\to1} > 0$, and suppose that for every $k \geq 0$*

$$\Delta_k \leq 2U_\gamma r_k, \qquad \frac{\gamma}{2X_\gamma \|A\|_{1\to1}^2} r_k^2 \leq \Delta_k - \Delta_{k+1}.$$

Then, for every $k \geq 1$,

$$0 \leq \Delta_k \leq \frac{8X_\gamma {U_\gamma}^2 \|A\|_{1\to1}^2}{\gamma} \frac{1}{k}. \tag{3}$$

For the cyclic KL iterates, Lemmas A.1 and A.2 supply these two scalar hypotheses with r_k equal to the finite block residual.

Proof sketch. The proof keeps the primal and dual viewpoints coupled. First, one half-sweep of block ascent enforces one block constraint exactly, so the remaining residual can be converted into a quantitative increase of F_γ by strong convexity of the KL geometry on the current primal mass shell. This gives a per-step ascent bound in terms of the block residuals, with constants depending only on γ, X_γ, and $\|A\|_{1\to1}$. Second, the dual gap is compared to the same residuals by testing the concave objective along a segment from the current dual variable to an optimal dual representative chosen in the quotient class; this is where the block-quotient radius U_γ enters. Combining the two estimates yields a nonlinear descent recursion $\Delta_k - \Delta_{k+1} \geq \alpha \Delta_k^2$ with $\alpha = \gamma/(8X_\gamma U_\gamma^2 \|A\|_{1\to1}^2)$, and summing the reciprocal gaps gives (3). The complete proof is given in Appendix A.

Numerical complexity of approximating the linear program $(\mathcal{P}_0)$. We now turn the dual gap estimate of Theorem 3.1 into a practical stopping rule for the unregularised programme $(\mathcal{P}_0)$. The next theorem combines the robust dual $O(1/k)$ rate with the standard entropic approximation tradeoff, choosing the temperature γ so that regularization and optimization errors share the target accuracy budget. The resulting guarantee uses only the primal and dual bounds X_γ, U_γ.

Theorem 3.2 (Accuracy versus runtime from primal/dual bounds)**.** *Let I be a finite nonempty coordinate set, let $C, x_0, x_\gamma \in \mathbb{R}^I$, let* Feas *be a feasible-set predicate, and let* $\mathrm{KL}(x) = \sum_{i \in I} \ell_i(x)$ *be the finite KL coordinate sum. Assume $\varepsilon > 0$ and a finite approximation certificate consisting of: the Section 3 rate certificate of Theorem 3.1; the temperature choice*

$$\gamma = \frac{\varepsilon}{2X_0^\star \log|I|}, \qquad 0 < X_0^\star \log|I|, \qquad X_0^\star \leq \mathrm{M}_{\max};$$

the finite KL-bias certificate of Lemma B.1 for $(C, x_0, x_\gamma, \mathrm{Feas}, \ell_i)$; and the identity

$$\Delta_{n+1} = \langle C, x_\gamma \rangle + \gamma \mathrm{KL}(x_\gamma) - F_\gamma^{(n)} \quad \textit{for every } n \geq 0.$$

If

$$\left\lceil 64 X_\gamma U_\gamma^2 \|A\|_{1\to 1}^2 \mathrm{M}_{\max} \frac{\log|I|}{\varepsilon^2} \right\rceil \leq n + 1,$$

then

$$\left| \langle C, x_0 \rangle - F_\gamma^{(n)} \right| \leq \varepsilon.$$

Proof sketch. The estimate is obtained by separating approximation and optimization errors. The regularization bias compares the unregularized optimum with the entropic optimum evaluated at the same feasible point; since the KL penalty is at most $X_0^\star \log d$ on a feasible point with mass $X_0^\star$ after normalization against the positive reference, the choice $\gamma = \varepsilon/(2X_0^\star \log d)$ spends half of the budget on bias. The remaining half is assigned to the dual optimization gap. Inserting this value of γ into Theorem 3.1 and requiring the displayed lower bound on k makes the robust $O(1/(\gamma k))$ term at most $\varepsilon/2$. The dual value, therefore, approximates the original LP value within ε. The detailed proof is in Appendix B.

Beyond KL. The same blueprint extends to general Bregman divergences under a generalized Pinsker condition. Appendix C states the abstract condition and the resulting rate. This extension covers objectives beyond linear programs, including semidefinite programs; quantum optimal transport is one example where the non-commutative Bregman geometry is natural.

4 Primal and Dual Bound Helpers

Our main result, Theorem 3.1, relies on uniform primal and dual bounds, denoted respectively by $X_{\max}$ and $U_{\max}$. We present here two helps to achieve this, shown in the blueprint of Figure 1 as the cyan and yellow boxes. The complete proofs of both helper results are in Appendix D.

Bounding Dual Iterates using Non-expansiveness of Ψ. The most difficult part is controlling the dual iterations. This section defines the geometric constants that govern this control and provides a generic blueprint that leverages the non-expansiveness of Ψ. It requires a primal bound H_γ (measuring deviation from the reference z in dual coordinates) and a control of the conditioning of the splitting through a decomposition constant κ.

Definition 4.1 (Primal bound H_γ in dual coordinates)**.** *We define $H_\gamma \in [0, +\infty]$ so that the minimizer x_γ of $(\mathcal{P}_\gamma)$ satisfies $\|\log x_\gamma - \log z\|_\infty \leq H_\gamma$.*

Definition 4.2 (Decomposition constant κ)**.** *The decomposition constant $\kappa = \kappa(A_1, A_2)$ is*

$$\kappa(A_1, A_2) := \sup_{y \in \mathrm{range}(A^\top),\, y \neq 0} \inf \left\{ \frac{\|w_1\|_\infty}{\|y\|_\infty} : \exists w_2 \text{ s.t. } A_1^\top w_1 + A_2^\top w_2 = y \right\} \in [0, +\infty]. \tag{4}$$

Proposition 4.1 (Uniform V_1-bound for alternating maximization)**.** *Let p be the seminorm used on the first dual block; in the applications below $p = \|\cdot\|_{V_1}$. Assume that the sweep map Ψ is non-expansive for p, i.e. $p(\Psi(a) - \Psi(b)) \leq p(a - b)$ for all a, b. Let $u_\star$ be a fixed point of Ψ and assume that its certified fixed-point budget is*

$$p(u_\star) \leq B.$$

For an arbitrary initialization $u_1^{(0)}$, define the first-block orbit by $u_1^{(k)} = \Psi^k(u_1^{(0)})$. Then, for all $k \geq 0$, $p\big(u_1^{(k)}\big) \leq p\big(u_1^{(0)}\big) + 2B$. In the KL-projection applications one uses the concrete budget $B = \kappa(\|C\|_\infty + \gamma H_\gamma)$.

Proof sketch. The non-expansiveness assumption turns the fixed point of the sweep into an orbit center: $p(u_1^{(k)}) \le p(u_1^{(0)}) + 2p(u_\star)$. The displayed fixed-point budget then gives the result. In the KL-projection applications, the budget $B = \kappa(\|C\|_\infty + \gamma H_\gamma)$ is obtained from optimality: the relation $x_\gamma = z \odot \exp((A^\top u_\gamma - C)/\gamma)$ implies that $A^\top u_\gamma$ is uniformly bounded by $\|C\|_\infty + \gamma H_\gamma$ after choosing a suitable representative. The decomposition constant κ converts this bound on $A^\top u_\gamma$ into a bound on the first block potential modulo the kernel. See Appendix D for the complete argument.

Bounding Primal Iterates from Dual Iterates. In many cases, one directly has access to a primal bound X_γ (for instance, in classical OT, $X_\gamma = 1$). If this is not the case, the following proposition shows that it is always possible to derive a bound X_γ from U_γ, with the issue being that it blows as $X_\gamma \sim \|b\|_1 U_\gamma/\gamma$ when $\gamma \to 0$. The main workload is thus to bound the dual potential in the block quotient semi-norm, which needs to be done on a case-by-case basis and exploit the structure of the split (A_1, A_2).

Proposition 4.2 (Primal bound from a dual bound)**.** *Let I be a finite coordinate set, let J be a finite potential set, and let $x_{\mathrm{mass}}^{(k)}$ denote the displayed primal mass associated with potentials $u^{(k)} \in \mathbb{R}^J$. Assume $\gamma > 0$, $C_{\min} \le C_i$ for every $i \in I$, and the following finite certificates: for every k there is a gauge shift $\sigma^{(k)} \in \mathbb{R}^J$ with*

$$\sum_{j \in J} b_j \sigma_j^{(k)} = 0, \qquad |u_j^{(k)} + \sigma_j^{(k)}| \le U_\gamma \quad \forall j \in J;$$

the displayed finite dual quantity is nondecreasing,

$$\sum_{j \in J} b_j u_j^{(k)} - \gamma x_{\mathrm{mass}}^{(k)} \le \sum_{j \in J} b_j u_j^{(k+1)} - \gamma x_{\mathrm{mass}}^{(k+1)};$$

and the zero start satisfies $u^{(0)} = 0$ and $x_{\mathrm{mass}}^{(0)} = \sum_{i \in I} \exp(-C_i/\gamma)$. Then for every $k \ge 0$,

$$x_{\mathrm{mass}}^{(k)} \le \frac{\sum_{j \in J} |b_j|}{\gamma} U_\gamma + |I| \, e^{-C_{\min}/\gamma}.$$

5 Sinkhorn–flow algorithm for W_1 on graphs

This section presents a new algorithm for approximating W_1 on graphs and demonstrates how to instantiate the general KL-projection blueprint. In Figure 1, this application is the green box, which enables the use of the primal and dual helpers to obtain robust rates. The section ends with numerical diagnostics on synthetic and genomic-inspired sparse graphs. Benchmarks are executed on CPU for reproducibility, and the same benchmark code can also be run on GPU.

W_1 distance on graphs. We consider an undirected graph with a vertex set $V = \{1, \dots, n\}$ and edge–length matrix $W \in \overline{\mathbb{R}}_+^{n \times n}$ ($\overline{\mathbb{R}}_+ := \mathbb{R}_+ \cup \{\infty\}$). An entry $W_{i,j} < \infty$ indicates the presence of the edge (i, j) with length $W_{i,j}$; we impose symmetry $W = W^\top$. Let $E = \{(i, j) \in V^2 : W_{i,j} < \infty\}$ and $p := |E|$ be the edge set and its cardinality. We denote $D \in \mathbb{R}_+^{n \times n}$ the shortest–path matrix, so that $D_{i,j}$ is the geodesic distance between vertices i and j. The graph is assumed to be connected, so that D is finite. As recalled in Appendix E, for any cost matrix C (for instance $C = D$, the geodesic distance), the transport distance W_C is defined by the Kantorovich linear program. We consider two probability vectors b_1, b_2 of size $m_1 = m_2 = n$ and take $C = D$. The resulting Optimal Transport distance $\mathrm{W}_D(b_1, b_2)$ is the Wasserstein-1 (W_1) distance, and it enjoys an alternate linear programming formulation which leverages the sparsity of the graph adjacency matrix W, the so-called Beckmann formulation [Beckmann, 1952, Santambrogio, 2015]. We introduce *flow* sparse matrices $f \in \mathbb{F} := \{f \in \mathbb{R}_+^{n \times n} \ : \ \forall (i, j) \notin E, f_{i,j} = 0\}$ whose entry $f_{i,j}$ encodes the amount transported *from j to i*. Note that $\mathbb{F}$ has dimensionality p (the set of edges). Define the discrete divergence operator $\mathrm{div}(f) := f^\top \mathbf{1}_n - f \mathbf{1}_n \in \mathbb{R}^n$. Beckmann's theorem yields the equivalent linear program

$$\mathrm{W}_D(b_1, b_2) = \min_{f \in \mathbb{F}} \{\langle W, f \rangle \ : \ \mathrm{div}(f) = b_1 - b_2\} \,. \tag{5}$$

Note that feasible f has non–zero entries only on E; the true variable dimension is therefore p rather than n^2. In practice, we store flows as sparse edge lists, and all arithmetic counts in Section 5 are expressed in terms of p.

Constraint splitting and the *flow–Sinkhorn* Algorithm. We duplicate the flow variable so that each affine block is easily projected onto. Write $x := (f, g) \in \mathbb{F}^2$ and $d = 2p$, where we recall $p = |E|$ is the number of finite entries of W and that flows are sparse matrices with p non-zero elements. With these lifted variables, the primal problem (5) can be re-written as

$$\mathrm{W}_D(b_1, b_2) = \min_{(f,g)\in\mathcal{C}_1\cap\mathcal{C}_2} \langle W, f\rangle + \langle W, g\rangle\,. \tag{6}$$

where the constraints can be written as $\mathcal{C}_1 := \{(f, g) : -f\mathbb{1}_n + g^\top\mathbb{1}_n = b_1 - b_2\}, \mathcal{C}_2 := \{(f, g) : f = g\}$. We then consider the entropic regularization $(\mathcal{P}_\gamma)$ with a fixed reference vector $z = (z_{i,j})_{i,j}$ supported on the edge $(i, j) \in E$

$$\min_{(f,g)\in\mathcal{C}_1\cap\mathcal{C}_2} \langle W, f\rangle + \langle W, g\rangle + \gamma\mathrm{KL}(f|z) + \gamma\mathrm{KL}(g|z).$$

Primal KL projections. Our Sinkhorn-flow algorithm is obtained by applying the iterative KL projection to solve this regularized problem. The KL projection on the two constraints can be computed explicitly as stated in the following proposition.

Proposition 5.1 (Closed–form KL projections)**.** *Let $b = b_1 - b_2$, let $h \geq 0$ have positive row and column sums $h_{\mathrm{row}}, h_{\mathrm{col}} \in \mathbb{R}^n_{++}$, and assume the finite variational certificate that the candidates below are the KL minimizers over the nonnegative flow cone for $\mathcal{C}_1$ and $\mathcal{C}_2$. Then*

$$\mathrm{Proj}_{\mathcal{C}_1}(h, h) = (\mathrm{diag}(s)\, h,\ h\ \mathrm{diag}(s)^{-1}), \qquad \mathrm{Proj}_{\mathcal{C}_2}(f, g) = (\sqrt{f \odot g},\ \sqrt{f \odot g}),$$

where

$$s_i = \phi\Big(\tfrac{b_i}{(h_{\mathrm{row}})_i}, \tfrac{(h_{\mathrm{col}})_i}{(h_{\mathrm{row}})_i}\Big) \in \mathbb{R}_{++}, \qquad \phi(t, u) := \frac{\sqrt{t^2 + 4u} - t}{2}.$$

Moreover the displayed $\mathcal{C}_1$ candidate is nonnegative, satisfies the divergence constraint, and obeys $s_i^2(h_{\mathrm{row}})_i + s_i b_i - (h_{\mathrm{col}})_i = 0$; the $\mathcal{C}_2$ candidate is nonnegative and satisfies $(\sqrt{f \odot g}_{ij})^2 = f_{ij}g_{ij}$.

Flow-Sinkhorn algorithm. We call the cyclic KL scheme of Section 2 applied to the lifted variable $x = (f, g)$ the *flow–Sinkhorn* algorithm. Since $\mathrm{Proj}_{\mathcal{C}_1}$ maps on pairs of equal flows, we track the output of this projection which simplifies the algorithmic description by focussing on a single flow variable and a single dual variable. The flow-Sinkhorn algorithm generates flows $f^{(k)}$ converging to the solution of the entropic regularization of the initial linear program (5).

$$(f^{(k+1)}, f^{(k+1)}) = \mathrm{Proj}_{\mathcal{C}_2} \circ \mathrm{Proj}_{\mathcal{C}_1}(f^{(k)}, f^{(k)}) \tag{7}$$

We denote $v^{(k)} \in \mathbb{R}^n$ the dual variable at iteration k of the algorithm and $s^{(k)} = e^{v^{(k)}/(2\gamma)}$ the associated scaling variable, which satisfies

$$f^{(k)}_{i,j} = z^C_{i,j}\, e^{\frac{v_i^{(k)} - v_j^{(k)}}{2\gamma}}, \quad f^{(k)} = \mathrm{diag}(s^{(k)}) z^C\, \mathrm{diag}(1/s^{(k)}), \tag{8}$$

where we set $z^C_{i,j} := z_{i,j}e^{-\frac{W_{i,j}}{\gamma}} \in \mathbb{F}$ (with the convention that $z^C_{i,j} = 0$ when there is no edge, i.e. $W_{i,j} = +\infty$). Proposition 5.2 gives closed forms for iterations (7) over these dual variables. The update of the dual is written in a stable way, so that formulas do not blow up numerically when $\gamma \to 0$, leveraging the stable implementation of the log-sum-exp operator $\mathcal{L}_\gamma(s) := \gamma \log\sum_j \exp(s_j/\gamma)$, implemented as $\mathcal{L}_\gamma(s - \max s) + \max s$. This stable formula is crucial in practice when targeting small regularization. Note that, in contrast with the usual Sinkhorn algorithm where $z^C_{i,j} > 0$, the Gibbs kernel in our setting may have vanishing entries due to the sparsity of the graph. In this case, the classical Sinkhorn algorithm is no longer guaranteed to converge linearly, because the Hilbert metric may fail to be contractive. We emphasize, however, that our algorithm differs from the classical Sinkhorn method, and our analysis does not rely on linear convergence rates.

Proposition 5.2 (Flow–Sinkhorn update in scaling variables)**.** *Let Ψ_1, Ψ_2 be the two block maps and set $\Psi = \Psi_1 \circ \Psi_2$. For $\gamma \neq 0$, assume that the second block produces the intermediate log-domain update*

$$m_i(v) = \frac{\alpha_i^-(v) - \alpha_i^+(v)}{2\gamma} + \mathrm{arsinh}(\beta_i(v)),$$

and that the first block satisfies $\Psi_1(m(v))_i = v_i/2 - \gamma m_i(v)$. Then the composed dual update is

$$\Psi(v)_i = \frac{1}{2}\, v_i\ +\ \frac{1}{2}\big(\alpha_i^+(v) - \alpha_i^-(v)\big)\ -\ \gamma\, \mathrm{arsinh}(\beta_i(v)), \quad \alpha_i^\pm(v) := \mathcal{L}_\gamma(-w_{i,\cdot} \pm v/2), \tag{9}$$

where $\mathrm{arsinh}(m) := \log(\sqrt{1 + m^2} + m)$ *and* $\beta_i(v) := \frac{b_{1,i} - b_{2,i}}{2}\, e^{-\frac{\alpha_i^+(v) + \alpha_i^-(v)}{2\gamma}}$.

Convergence Analysis. The following theorem establishes dual convergence for the graph-flow instantiation by plugging the graph-specific helper bounds into the general theorem.

Theorem 5.1 (Sinkhorn–flow complexity)**.** *Fix an iterate k and assume the following operation-budget certificates: $\varepsilon > 0$, $p \geq 0$, W_1 error at iterate k is at most ε, $k \leq K_{\text{iter}}$,*

$$K_{\text{iter}} \leq L\, \frac{\text{diameter}(E)^3}{\varepsilon^4}, \qquad \text{ops}_{\text{sweep}} \leq p, \qquad \text{ops}_{\text{total}} \leq k\, \text{ops}_{\text{sweep}},$$

and the edge-count family satisfies $p = p(\varepsilon)$ with $p(\varepsilon) = o(1/\log(1/\varepsilon))$ as $\varepsilon \downarrow 0$. Then Sinkhorn–flow has ε-additive error at iterate k and

$$\text{ops}_{\text{total}} \leq L\, p\, \frac{\text{diameter}(E)^3}{\varepsilon^4},$$

up to the logarithmic factors absorbed in L.

Proof sketch. The proof instantiates the green box of Figure 1. The variables are $x = (f, g) \in \mathbb{F}^2$, the two constraint blocks are the divergence constraint $A_1(f, g) = b_1 - b_2$ and the equality constraint $A_2(f, g) = 0$, and the dual variables are $u = (v, U)$, where $A_1(f, g) = f\mathbb{1}_n - g^\top \mathbb{1}_n \in \mathbb{R}^n$ and $A_2(f, g) = f - g \in \mathbb{F}$. Appendix F identifies the quotient norms as variation seminorms and proves the signed non-expansiveness needed by the dual helper: the signature is $\Sigma = \text{diag}(+I_E, -I_E)$, the translation parameter is $\tau = +1$, and the second block update satisfies $\Psi_2(v)_{i,j} = (v_j - v_i)/2$ with $\|\Psi_2(v)\|_{V_2} \leq \|v\|_{V_1}$. The same appendix proves the graph decomposition estimate $\kappa \leq 2\,\text{diameter}(E)$ and the explicit bound $|A|_{1\to 1} = \|A\|_{1\to 1} = 2$ for this split. The dual helper therefore gives $U_\gamma = O(\text{diameter}(E)(W_{\max} + \gamma H_\gamma))$. The graph H_γ estimate and the positive-cost lower bound then feed the primal helper, giving a mass bound $X_\gamma = O(\text{diameter}(E)/\gamma + pe^{-W_{\min}/\gamma})$ for probability inputs. Plugging these U_γ, X_γ, and $|A|_{1\to 1}$ values into Theorem 3.2, and taking $\gamma \asymp \varepsilon$, yields $O(\text{diameter}(E)^3/\varepsilon^4)$ iterations up to logarithmic factors. Each sweep uses sparse edge operations, hence costs $O(p)$, which gives the stated arithmetic complexity.

Numerical experiments. The public implementation includes a PyTorch implementation of flow-Sinkhorn together with scripts reproducing these plots. We benchmark in the regime emphasized by our theory: approximation speed toward the *unregularized* solution as a function of wall-clock time and regularization γ, rather than raw per-iteration decay at fixed regularization geometry. The error plotted below is the best-so-far relative Euclidean error on the recovered graph flow, $\min_{s\leq t} \|f^{(s)} - f^\star\|_2/\|f^\star\|_2$, where $f^\star$ is the unregularized min-cost-flow solution and each method is evaluated through the corresponding graph-flow representation. Flow-Sinkhorn and vanilla Sinkhorn are therefore plotted together on the same axes (solid and dashed, respectively), with a common time horizon. This objective should, however, be interpreted with care for machine-learning practice: in many downstream tasks, keeping a non-vanishing (often larger) regularization can improve stability and generalization, so fastest convergence to the unregularized solution is not always the relevant end goal. The benchmark settings are:

- **Line graph.** Segment graph with nearest-neighbor connectivity, $n = 80$ nodes and two localized endpoint measures on the segment extremities; edge weights are unit lengths.
- **Delaunay sparse graphs.** Planar point cloud with $n = 140$ nodes, Delaunay triangulation connectivity, and Euclidean edge weights; localized source/target masses are sampled on spatially separated regions.
- **Single-cell sparse graphs.** We use the Waddington-OT single-cell RNA-seq data of Schiebinger et al. [2019], which follows mouse embryonic fibroblast reprogramming toward pluripotency. The displayed graph is built on a subset of $n = 240$ cells, sampled as 60 cells from each of the first four snapshots (days $0, 0.5, 1, 1.5$), embedded by PCA (dimension 30) and connected by a k-NN graph ($k = 4$). The blue and red measures are the empirical distributions at the initial and final sampled snapshots; the displayed flow is therefore a coarse transport proxy for developmental trajectories and fate progression across the sampled cell-state manifold.

For graph visualizations, we aggregate directed flows into undirected magnitudes $|f_{ij}| + |f_{ji}|$, draw all graph edges in thin gray, and overlay only upper-quantile transport edges in thicker orange; source/target supports are shown as blue/red markers. Figures 2 and 3 summarize the CPU runs (for the 3 settings) and the GPU (NVIDIA Quadro RTX 4000 (8GB)) runs (for the line graph only, but with 3 different sizes n). In the GPU runs, increasing the line graph size makes the separation between vanilla Sinkhorn and flow-Sinkhorn more pronounced: the sparse flow formulation keeps its edge-local arithmetic, while vanilla Sinkhorn still acts on the dense path metric.

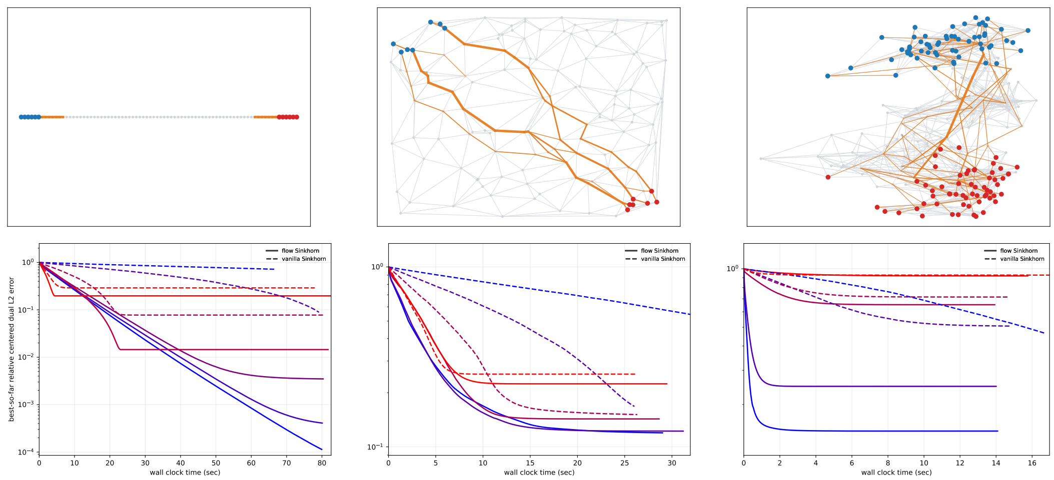

Figure 2: Top row: benched graph together with the flow f (line graph, planar Delaunay, single cell). Bottom row: ℓ^2 error $\|f^{(s)} - f^\star\|_2 / \|f^\star\|_2$, against wall-clock time. Flow Sinkhorn is plain and vanilla Sinkhorn is dashed; colors encode γ from blue (small) to red (large). Gamma ranges are: line (flow $\gamma \in [10^{-3}, 5]$, vanilla $\gamma \in [10^{-1}, 1]$), Delaunay (flow $\gamma \in [9 \times 10^{-4}, 9 \times 10^{-3}]$, vanilla $\gamma \in [9 \times 10^{-4}, 9 \times 10^{-3}]$), single-cell (flow $\gamma \in [3 \times 10^{-1}, 9]$, vanilla $\gamma \in [1, 5]$).

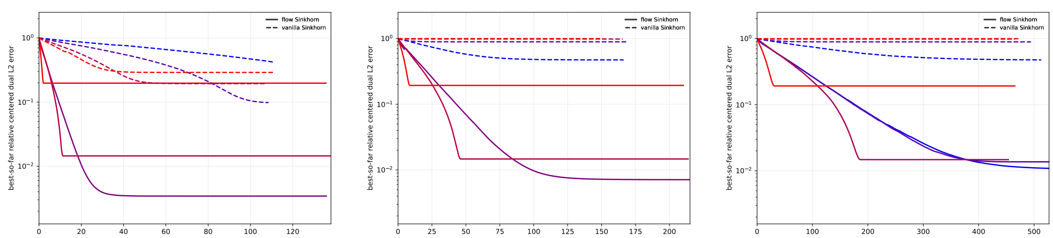

Figure 3: GPU line-graph benchmark, same best-so-far relative ℓ^2 flow error as in Figure 2. From left to right: small ($n = 80, m = 6$), large ($n = 160, m = 12$), and larger ($n = 320, m = 24$) line-segment tests.

Conclusion

We have presented an analysis of iterative Bregman projection methods that yields sublinear convergence rates but with constants that scale favorably with the dimension. Such a regime is the relevant one for studying approximation rates in machine learning, where problems are typically obtained through sampling and discretization, and dimensional dependence plays a central role. While the behavior of these methods was well understood in the classical Optimal Transport setting, it was less clear which structural properties were truly responsible for their stability and effectiveness. This article identifies and isolates these pivotal ingredients, and proposes a generic blueprint that applies beyond the classical case. The relevance of the analysis is illustrated through its application to network flow formulations, leading to a cheap and easy-to-implement algorithm for computing the Wasserstein-1 distance on graphs.

References

Ravindra K. Ahuja, Thomas L. Magnanti, and James B. Orlin. *Network Flows: Theory, Algorithms, and Applications*. Prentice Hall, 1993.

Jason Altschuler, Jonathan Niles-Weed, and Philippe Rigollet. Near-linear time approximation algorithms for optimal transport via Sinkhorn iteration. *Advances in Neural Information Processing Systems*, 30, 2017.

Pierre-Cyril Aubin-Frankowski, Anna Korba, and Flavien Léger. Mirror descent with relative smoothness in measure spaces, with application to Sinkhorn and EM. *Advances in Neural Information Processing Systems*, 35:17263–17275, 2022.

Martin Beckmann. A continuous model of transportation. *Econometrica*, 20:643–660, 1952.

Jean-David Benamou, Guillaume Carlier, Marco Cuturi, Luca Nenna, and Gabriel Peyré. Iterative bregman projections for regularized transportation problems. *SIAM Journal on Scientific Computing*, 37(2):A1111–A1138, 2015.

Jonathan Borwein, Adrian Lewis, and Roger Nussbaum. Entropy minimization, dad problems, and doubly stochastic kernels. *Journal of Functional Analysis*, 123(2):264–307, 1994.

Lev M. Bregman. The relaxation method of finding the common point of convex sets and its application to the solution of problems in convex programming. *USSR Computational Mathematics and Mathematical Physics*, 7(3):200–217, 1967.

Emanuele Caglioti, François Golse, and Thierry Paul. Quantum optimal transport is cheaper. *Journal of Statistical Physics*, 181(1):149–162, 2020. doi: 10.1007/s10955-020-02571-7.

Michael Candrall and Luc Tartar. Some relations between non expansive and order preserving maps. *Proceedings of the AMS*, 78(3):385–390, 1980.

Guillaume Carlier. On the linear convergence of the multimarginal sinkhorn algorithm. *SIAM Journal on Optimization*, 32(2):786–794, 2022.

Guillaume Carlier and Filippo Santambrogio. A continuous theory of traffic congestion and wardrop equilibria. *Journal of Mathematical Sciences*, 181(6):792–804, 2012.

Yair Censor and Michal Rezač. Iterative projection methods in structured optimization. *Optimization*, 64(11):2343–2361, 2015.

Shouvanik Chakrabarti, Yiming Huang, Tongyang Li, Soheil Feizi, and Xiaodi Wu. Quantum wasserstein generative adversarial networks. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

Deeparnab Chakrabarty and Sanjeev Khanna. Better and simpler error analysis of the sinkhorn–knopp algorithm for matrix scaling. *Mathematical Programming*, 188(1):395–407, 2021.

Li Chen, Rasmus Kyng, Yang Liu, Richard Peng, Maximilian Probst Gutenberg, and Sushant Sachdeva. Maximum flow and minimum-cost flow in almost-linear time. *Journal of the ACM*, 72 (3):1–103, 2025.

Yongxin Chen, Tryphon Georgiou, and Michele Pavon. Entropic and displacement interpolation: A computational approach using the hilbert metric. *SIAM Journal on Applied Mathematics*, 76(6): 2375–2396, 2016.

Lénaïc Chizat, Gabriel Peyré, Bernhard Schmitzer, and François-Xavier Vialard. Scaling algorithms for unbalanced transport problems. *Mathematics of Computation*, 87(314):2563–2609, 2018.

Lénaïc Chizat, Pierre Roussillon, Flavien Léger, François-Xavier Vialard, and Gabriel Peyré. Faster Wasserstein distance estimation with the Sinkhorn divergence. *Advances in Neural Information Processing Systems*, 33:2257–2269, 2020.

Lénaïc Chizat, Alex Delalande, and Tomas Vaškevičius. Sharper exponential convergence rates for sinkhorn's algorithm in continuous settings. *Mathematical Programming*, pages 1–50, 2025.

Giovanni Conforti, Alain Durmus, and Giacomo Greco. Quantitative contraction rates for Sinkhorn algorithm: beyond bounded costs and compact marginals. *arXiv preprint arXiv:2304.04451*, 2023.

Imre Csiszár and Gábor Tusnády. Information geometry and alternating minimization procedures. *Statistics & Decisions*, Supplement 1:205–237, 1984.

Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems*, volume 26, pages 2292–2300, 2013.

Samuel I Daitch and Daniel A Spielman. Faster approximate lossy generalized flow via interior point algorithms. In *Proceedings of the fortieth annual ACM symposium on Theory of computing*, pages 451–460, 2008.

George Deligiannidis, Valentin de Bortoli, and Arnaud Doucet. Quantitative uniform stability of the iterative proportional fitting procedure. *The Annals of Applied Probability*, 34(1A):501–516, 2024.

William Edwards Deming and Frederick F. Stephan. On a least squares adjustment of a sampled frequency table when the expected marginal totals are known. *The Annals of Mathematical Statistics*, 11(4):427–444, 1940.

Sally Dong, Yu Gao, Gramoz Goranci, Yin Tat Lee, Sushant Sachdeva, Richard Peng, and Guanghao Ye. Nested dissection meets ipms: Planar min-cost flow in nearly-linear time. *Journal of the ACM*, 72(4):1–75, 2025.

Pavel Dvurechensky, Alexander Gasnikov, and Alexey Kroshnin. Computational optimal transport: Complexity by accelerated gradient descent is better than by Sinkhorn's algorithm. In *International conference on machine learning*, pages 1367–1376, 2018.

Stephan Eckstein. Hilbert's projective metric for functions of bounded growth and exponential convergence of sinkhorn's algorithm, 2025.

Steven N Evans and Frederick A Matsen. The phylogenetic Kantorovich–Rubinstein metric for environmental sequence samples. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 74(3):569–592, 2012.

Joel N. Franklin and Jens Lorenz. On the scaling of multidimensional matrices. *Linear Algebra and Its Applications*, 114–115:717–735, 1989.

Kurt Friedrichs. On certain inequalities and characteristic symmetric bilinear forms. *Mathematische Annalen*, 115:249–290, 1938.

François Golse, Clément Mouhot, and Thierry Paul. On the mean field and classical limits of quantum mechanics. *Communications in Mathematical Physics*, 343(1):165–205, 2016. doi: 10.1007/s00220-015-2485-7.

Kristen Grauman and Trevor Darrell. Fast contour matching using approximate earth mover's distance. In *Proceedings of the 2004 IEEE Computer Society Conference on Computer Vision and Pattern Recognition, 2004. CVPR 2004.*, volume 1, pages I–I. IEEE, 2004.

Giacomo Greco, Maxence Noble, Giovanni Conforti, and Alain Durmus. Non-asymptotic convergence bounds for sinkhorn iterates and their gradients: a coupling approach. In *Proceedings of Thirty Sixth Conference on Learning Theory*, volume 195, pages 716–746, 2023.

Fumio Hiai, Masanori Ohya, and Makoto Tsukada. Sufficiency, KMS condition and relative entropy in von Neumann algebras. *Pacific Journal of Mathematics*, 96(1):99–109, 1981. doi: 10.2140/pjm.1981.96.99.

Bahman Kalantari, Isabella Lari, Federica Ricca, and Bruno Simeone. On the complexity of general matrix scaling and entropy minimization via the RAS algorithm. *Mathematical Programming*, 112 (2):371–401, 2008.

Matt Kusner, Yu Sun, Nicholas Kolkin, and Kilian Weinberger. From word embeddings to document distances. In *International conference on machine learning*, pages 957–966. PMLR, 2015.

Flavien Léger. A gradient descent perspective on Sinkhorn. *Applied Mathematics & Optimization*, 84(2):1843–1855, 2021.

Simone Di Marino and Augusto Gerolin. An optimal transport approach for the schrödinger bridge problem and convergence of sinkhorn algorithm. *Journal of Scientific Computing*, 85(2):27, 2020.

Lipeng Ning, Tryphon T. Georgiou, and Allen Tannenbaum. On matrix-valued monge–kantorovich optimal mass transport. *IEEE Transactions on Automatic Control*, 60(2):373–382, 2015. doi: 10.1109/TAC.2014.2350171.

Gabriel Peyré, Lénaïc Chizat, François-Xavier Vialard, and Justin Solomon. Quantum entropic regularization of matrix-valued optimal transport. *European Journal of Applied Mathematics*, 30 (6):1079–1102, 2019. doi: 10.1017/S0956792517000274.

Yossi Rubner, Carlo Tomasi, and Leonidas J Guibas. A metric for distributions with applications to image databases. In *Sixth international conference on computer vision (IEEE Cat. No. 98CH36271)*, pages 59–66. IEEE, 1998.

Romeil Sandhu, Tryphon Georgiou, Ed Reznik, Liangjia Zhu, Ivan Kolesov, Yasin Senbabaoglu, and Allen Tannenbaum. Graph curvature for differentiating cancer networks. *Scientific reports*, 5(1): 12323, 2015.

Filippo Santambrogio. *Optimal transport for applied mathematicians*. Springer, 2015.

Geoffrey Schiebinger, Jian Shu, Marcin Tabaka, Brian Cleary, Vidya Subramanian, Aryeh Solomon, Samuel Liu, Stacie Lin, Paul Berube, Lance Lee, et al. Optimal-transport analysis of single-cell gene expression identifies developmental trajectories in reprogramming. *Cell*, 176(4):928–943, 2019.

Jayson Sia, Edmond Jonckheere, and Paul Bogdan. Ollivier-Ricci curvature-based method to community detection in complex networks. *Scientific reports*, 9(1):9800, 2019.

R. Sinkhorn. A relationship between arbitrary positive matrices and doubly stochastic matrices. *Ann. Math. Statist.*, 35:876–879, 1964.

Justin Solomon, Raif Rustamov, Leonidas Guibas, and Adrian Butscher. Earth mover's distances on discrete surfaces. *ACM Transactions on Graphics (ToG)*, 33(4):1–12, 2014.

Armin Uhlmann. Relative entropy and the Wigner–Yanase–Dyson–Lieb concavity in an interpolation theory. *Communications in Mathematical Physics*, 54(1):21–32, 1977. doi: 10.1007/BF01609834.

John Watrous. *The Theory of Quantum Information*. Cambridge University Press, 2018. doi: 10.1017/9781316848142.

Udny Yule. On the methods of measuring association between two attributes. *Journal of the Royal Statistical Society*, 75(6):579–652, 1912.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes].

   Justification: The claims are the robust dual convergence blueprint, its OT and graph-W_1 instantiations, supplementary benchmarks, and Lean formalization.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes].

   Justification: The assumptions needed for the blueprint are explicit, and the beyond-KL appendix describes open issues for quantum OT.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete proof?

Answer: [Yes].

Justification: The main text states assumptions and sketches the main proofs; complete proofs are provided in the appendix, with Lean certification for the main blueprint chain.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

   Answer: [Yes].

   Justification: Benchmarking is provided through the public repository, together with the PyTorch implementation and benchmarking scripts.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to reproduce the results, as described in supplemental material?

   Answer: [Yes].

   Justification: The public repository contains code, benchmark scripts, and the Lean formalization.

6. **Experimental setting/details**

   Question: Does the paper specify the details necessary to understand the benchmark results?

   Answer: [Yes].

   Justification: Section 5 describes the common methodology and the benchmark settings, with implementation details in supplementary scripts.

7. **Experiment statistical significance**

   Question: Does the paper report error bars or other appropriate statistical information?

   Answer: [N/A].

   Justification: The benchmarks are deterministic solver diagnostics rather than stochastic comparisons supporting statistical claims.

8. **Experiments compute resources**

   Question: Does the paper provide sufficient information on compute resources needed to reproduce the experiments?

   Answer: [Yes].

   Justification: The theoretical per-iteration cost is stated, and the supplementary benchmarking scripts document the implementation path.

9. **Code of ethics**

   Question: Does the research conform to the NeurIPS Code of Ethics?

   Answer: [Yes].

   Justification: This theoretical optimization work does not involve human subjects, data collection, or deployed models.

10. **Broader impacts**

    Question: Does the paper discuss potential positive and negative societal impacts?

    Answer: [N/A].

    Justification: The work is foundational optimization theory and has no direct deployment pathway requiring safeguards.

11. **Safeguards**

    Question: Does the paper describe safeguards for responsible release of high-risk data or models?

    Answer: [N/A].

Justification: The work releases mathematical code and formal proofs, not high-risk data or models.

12. **Licenses for existing assets**

    Question: Are existing assets properly credited and licensed?

    Answer: [N/A].

    Justification: The supplementary assets are code and formalization artifacts produced for this work.

13. **New assets**

    Question: Are new assets introduced in the paper well documented?

    Answer: [Yes].

    Justification: The public repository documents the code, benchmark scripts, and Lean formalization.

14. **Crowdsourcing and research with human subjects**

    Question: Does the paper include details for crowdsourcing or human-subject research?

    Answer: [N/A].

    Justification: The work does not use crowdsourcing or human participants.

15. **Institutional review board approvals**

    Question: Does the paper describe IRB approvals or equivalent for human-subject research?

    Answer: [N/A].

    Justification: The work does not involve human-subject research.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods?

    Answer: [Yes].

    Justification: LLMs were used to generate and check the Lean formalization of the proof. The informal mathematical proofs were completed and carefully checked by the authors; the Lean proofs are provided in the public repository as an additional machine-checkable certificate.

A Proof of the sublinear dual rate

Proof of Theorem 3.1.

Proof. The proof relies on a per-step ascent estimate (Lemma A.1) and a dual-gap–to–residual comparison (Lemma A.2). Write the global residual as $r^{(k)} = Ax^{(k)} - b$ and recall $X_\gamma \geq \|x^{(k)}\|_1$ for all k. Summing (A1) and (A2) in Lemma A.1 yields, for a full outer sweep, denoting $\lambda := \frac{\gamma}{2X_\gamma \|A\|^2_{1\to 1}}$

$$\Delta_k - \Delta_{k+1} \geq \lambda(\|r_1^{(k)}\|_1^2 + \|r_2^{(k+\frac{1}{2})}\|_1^2) \geq \lambda \|r_1^{(k)}\|_1^2.$$

By Lemma A.2, specialized to the first half-sweep with second residual block equal to zero, $\Delta_k \leq 2U_\gamma \|r_1^{(k)}\|_1$, hence $\|r_1^{(k)}\|_1 \geq \Delta_k/(2U_\gamma)$. Substituting into the previous inequality gives

$$\Delta_k - \Delta_{k+1} \geq \alpha \Delta_k^2, \quad \text{where} \quad \alpha := \frac{\lambda}{4{U_\gamma}^2}.$$

Dividing by $\Delta_k \Delta_{k+1}$ and using the fact that Δ_k is decaying gives $\frac{1}{\Delta_{k+1}} - \frac{1}{\Delta_k} \geq \alpha$. Summing from 0 to $k-1$ yields

$$\frac{1}{\Delta_k} \geq \frac{1}{\Delta_0} + \alpha k \geq \alpha k,$$

since $\Delta_0 > 0$. Therefore, for every $k \geq 1$, $\Delta_k \leq \frac{1}{\alpha k}$, which is exactly the estimate of Theorem 3.1. □

The proof of Theorem 3.1 rests on two ingredients: Lemma A.1 quantifies the ascent achieved during one half–sweep of the dual ascent the two block updates; Lemma A.2 relates the resulting dual gap to the primal residuals. In the following, let $u^{(k)}$ denote the dual iterates and $x^{(k)} := x(u^{(k)})$ the corresponding primal iterates. For the Lean formalization we use the following finite certificate form of one block update. Given finite vectors p, q, a common mass M, a temperature γ, and two dual objective values F^-, F^+, write

$$\mathsf{ExactBlockCert}_{M,\gamma}(p, q; F^-, F^+)$$

when $p, q \geq 0$, $\sum_i p_i = \sum_i q_i = M$, $q_i = 0 \Rightarrow p_i = 0$, and

$$F^+ = F^- + \gamma \sum_i p_i \log \frac{p_i}{q_i}.$$

The support condition is the finite absolute-continuity condition needed to state the KL term on the boundary of the simplex. In the concrete KL projection half-step, p and q are respectively the primal vector after and before the block projection, so that the displayed increment is the usual $\gamma\mathrm{KL}(\text{after}\|\text{before})$ gain.

Lemma A.1 (Per–step ascent for the dual blocks)**.** *Let $F_k := F_\gamma(u_1^{(k)}, u_2^{(k)})$ and $F_{k+\frac{1}{2}} := F_\gamma(u_1^{(k+1)}, u_2^{(k)})$. Assume $\gamma \geq 0$, $X_\gamma > 0$, $\|A\|_{1\to 1} > 0$, and that there is a common mass $M > 0$ with $M \leq X_\gamma$. For each k, let $(p_1^{(k)}, q_1^{(k)})$ and $(p_2^{(k)}, q_2^{(k)})$ be finite after/before vectors for the two half-steps, satisfying*

$$\mathsf{ExactBlockCert}_{M,\gamma}\big(p_1^{(k)}, q_1^{(k)}; F_k, F_{k+\frac{1}{2}}\big), \qquad \mathsf{ExactBlockCert}_{M,\gamma}\big(p_2^{(k)}, q_2^{(k)}; F_{k+\frac{1}{2}}, F_{k+1}\big).$$

Let $r_1^{(k)}, r_2^{(k)} \geq 0$ be residual proxies such that

$$r_1^{(k)} \leq \|A\|_{1\to 1}\, \|p_1^{(k)} - q_1^{(k)}\|_1, \qquad r_2^{(k)} \leq \|A\|_{1\to 1}\, \|p_2^{(k)} - q_2^{(k)}\|_1.$$

Then for every $k \geq 0$,

$$F_\gamma\big(u_1^{(k+1)}, u_2^{(k)}\big) - F_\gamma\big(u_1^{(k)}, u_2^{(k)}\big) \geq \frac{\gamma}{2X_\gamma} \frac{\big(r_1^{(k)}\big)^2}{\|A\|_{1\to 1}^2}, \tag{A1}$$

$$F_\gamma\big(u_1^{(k+1)}, u_2^{(k+1)}\big) - F_\gamma\big(u_1^{(k+1)}, u_2^{(k)}\big) \geq \frac{\gamma}{2X_\gamma} \frac{\big(r_2^{(k)}\big)^2}{\|A\|_{1\to 1}^2}. \tag{A2}$$

Proof. We prove (A1); the argument for (A2) is identical with the roles of (A_1, u_1, r_1) and (A_2, u_2, r_2) swapped. The first certificate says, with $p = p_1^{(k)}$ and $q = q_1^{(k)}$, that the finite after/before vectors are nonnegative, have the same mass M, satisfy the support domination $q_i = 0 \Rightarrow p_i = 0$, and obey the exact gamma-scaled dual gain

$$F_\gamma\big(u_1^{(k+1)}, u_2^{(k)}\big) - F_\gamma\big(u_1^{(k)}, u_2^{(k)}\big) = \gamma\mathrm{KL}(p|q).$$

By the non–normalised Pinsker inequality, stated for completeness as Lemma A.3 below and applied on the common mass shell of the KL projection, $\mathrm{KL}(p|q) \geq \frac{\|p-q\|_1^2}{2\sum_i p_i}$. Since $\sum_i p_i = M \leq X_\gamma$, this gives

$$\mathrm{KL}(p|q) \geq \frac{\|p - q\|_1^2}{2X_\gamma}.$$

The residual proxy assumption gives

$$r_1^{(k)} \leq \|A\|_{1\to 1} \|p - q\|_1.$$

Because $r_1^{(k)} \geq 0$ and $\|A\|_{1\to 1} > 0$, squaring and substituting this estimate in the Pinsker lower bound yields (A1). In the concrete block update, this scalar proxy is instantiated by the ℓ^1 norm of the block residual, and the displayed inequality above is the usual operator-norm estimate. $\square$

If the dual iterates stay bounded, the following result relates the current dual gap to the ℓ^1 residuals. We state the finite two-block quotient estimate used in the Lean certificate. For a sign $\tau \in \{+1, -1\}$, set

$$\|(a,b)\|_{V,\tau} \coloneqq \inf_{c\in\mathbb{R}} \max\{\|a + c\mathbb{1}\|_\infty, \|b + \tau c\mathbb{1}\|_\infty\}, \qquad \|(r_1, r_2)\|_{1,1} \coloneqq \sum_i |(r_1)_i| + \sum_j |(r_2)_j|.$$

Lemma A.2 (Dual gap versus global residual)**.** *Let I_1, I_2 be finite nonempty index sets. Let $r_1 \in \mathbb{R}^{I_1}$, $r_2 \in \mathbb{R}^{I_2}$ and let $u^\star = (u_1^\star, u_2^\star)$ and $u = (u_1, u_2)$ be two dual block pairs. Assume the paired residual compatibility*

$$\sum_{i\in I_1} (r_1)_i + \tau \sum_{j\in I_2} (r_2)_j = 0,$$

and assume that the current gap proxy G satisfies the concavity-pairing bound

$$G \le \sum_{i\in I_1} (r_1)_i\big((u_1^\star)_i - (u_1)_i\big) + \sum_{j\in I_2} (r_2)_j\big((u_2^\star)_j - (u_2)_j\big).$$

If the two signed quotient radii satisfy

$$\|u^\star\|_{V,\tau} < U_\gamma, \qquad \|u\|_{V,\tau} < U_\gamma,$$

then

$$G \le 2U_\gamma \, \|(r_1, r_2)\|_{1,1}.$$

Proof. For any two scalar shifts $c^\star, c \in \mathbb{R}$, the compatibility assumption gives

$$\begin{aligned}
&\sum_i (r_1)_i\big((u_1^\star)_i - (u_1)_i\big) + \sum_j (r_2)_j\big((u_2^\star)_j - (u_2)_j\big) \\
&\quad = \sum_i (r_1)_i\big((u_1^\star)_i + c^\star - (u_1)_i - c\big) + \sum_j (r_2)_j\big((u_2^\star)_j + \tau c^\star - (u_2)_j - \tau c\big).
\end{aligned}$$

Applying Hölder blockwise, then using $|a - b| \le |a| + |b|$, gives

$$G \le \|(r_1, r_2)\|_{1,1}\Big(\max\{\|u_1^\star + c^\star\mathbb{1}\|_\infty, \|u_2^\star + \tau c^\star\mathbb{1}\|_\infty\} + \max\{\|u_1 + c\mathbb{1}\|_\infty, \|u_2 + \tau c\mathbb{1}\|_\infty\}\Big).$$

Since both signed quotient radii are strictly below U_γ, representatives can be chosen in the two signed-shift orbits with block sup-norms at most U_γ. Substituting those representatives yields $G \le 2U_\gamma\|(r_1, r_2)\|_{1,1}$. □

In the generated dual iterates, the first half-sweep has $r_2 = 0$ because the second constraint block is already satisfied, while the second half-sweep has the analogous first-block residual equal to zero. Applying Lemma A.2 to these two specializations gives

$$\Delta_k \le 2U_\gamma\|r_1^{(k)}\|_1, \quad \Delta_{k+1/2} \le 2U_\gamma\|r_2^{(k+1/2)}\|_1. \tag{10}$$

Proposition A.1 (Normalized Pinsker inequality)**.** *Let $\mu, \nu \in \mathbb{R}^d$ satisfy $\mu_i \ge 0$, $\nu_i > 0$ for all i and $\sum_i \mu_i = \sum_i \nu_i = 1$. On the probability shell we write $\mathrm{KL}(\mu|\nu) = \sum_i \mu_i \log(\mu_i/\nu_i)$, with the convention $0\log(0/\nu_i) = 0$. Then*

$$\mathrm{KL}(\mu|\nu) \ge \frac{1}{2}\|\mu - \nu\|_1^2.$$

Proof. Using the variational representation of relative entropy,

$$\mathrm{KL}(\mu|\nu) = \sup_{f\in\mathbb{R}^d} \left\{ \langle \mu, f\rangle - \log\Big(\sum_i \nu_i e^{f_i}\Big)\right\},$$

choose $f_i = \lambda s_i$ where $s_i = \mathrm{sign}(\mu_i - \nu_i) \in \{-1, 1\}$ and $\lambda \ge 0$. Then $\langle \mu - \nu, s\rangle = \|\mu - \nu\|_1$. Writing $a = \sum_{i:s_i=1} \nu_i$, $b = \sum_{i:s_i=-1} \nu_i$ gives $\sum_i \nu_i e^{\lambda s_i} = ae^{\lambda} + be^{-\lambda}$ with $a + b = 1$, hence

$$\log\Big(\sum_i \nu_i e^{\lambda s_i}\Big) \le \lambda \sum_i \nu_i s_i + \frac{\lambda^2}{2}.$$

Therefore

$$\mathrm{KL}(\mu|\nu) \ge \lambda\|\mu - \nu\|_1 - \frac{\lambda^2}{2} \qquad (\lambda \ge 0),$$

and choosing $\lambda = \|\mu - \nu\|_1$ yields the claim. □

Lemma A.3 (Non–normalised Pinsker inequality)**.** *Let $p, q \in \mathbb{R}^d$ satisfy $p_i \geq 0$, $q_i > 0$ for all i and have the same positive mass $M = \sum_i p_i = \sum_i q_i > 0$. On this common-mass shell, the full non-normalised divergence $\sum_i (p_i \log(p_i/q_i) - p_i + q_i)$ equals $\sum_i p_i \log(p_i/q_i)$; we use this common-mass form below. Then*

$$\mathrm{KL}(p|q) \geq \frac{\|p-q\|_1^2}{2M}.$$

Proof. Set $\bar{p} = p/M$ and $\bar{q} = q/M$. Then $\bar{p}, \bar{q} \in \Delta_d$, so $\bar{p}_i \geq 0$, $\bar{q}_i > 0$, $\sum_i \bar{p}_i = \sum_i \bar{q}_i = 1$, and $\mathrm{KL}(\bar{p}|\bar{q}) \geq \frac{1}{2}\|\bar{p} - \bar{q}\|_1^2$ by Proposition A.1. By homogeneity on a common mass shell, $\mathrm{KL}(p|q) = M\,\mathrm{KL}(\bar{p}|\bar{q})$ and $\|\bar{p} - \bar{q}\|_1 = M^{-1}\|p - q\|_1$. Therefore

$$\mathrm{KL}(p|q) = M\,\mathrm{KL}(\bar{p}|\bar{q}) \geq \frac{M}{2}\|\bar{p} - \bar{q}\|_1^2 = \frac{\|p-q\|_1^2}{2M},$$

as claimed. $\square$

B KL bias and runtime proof

Lemma B.1 (KL bias)**.** *Let I be a finite nonempty index set and set $d := |I|$. Let $C, x_0, x_\gamma : I \to \mathbb{R}$, let $\mathcal{F} \subseteq \mathbb{R}^I$ be a feasible set, and let $\ell_i : \mathbb{R}^I \to \mathbb{R}$ be coordinate KL terms. Define*

$$\mathrm{KL}(x) := \sum_{i \in I} \ell_i(x).$$

Write

$$L_C(x) := \sum_{i \in I} C_i x_i, \qquad R_\gamma(x) := L_C(x) + \gamma\,\mathrm{KL}(x).$$

Assume $\gamma \geq 0$, that x_0 minimizes L_C over $\mathcal{F}$, that x_γ minimizes R_γ over $\mathcal{F}$, and that $\mathrm{KL}(x) \geq 0$ for every $x \in \mathcal{F}$. Assume moreover that

$$\ell_i(x_0) \leq x_{0,i} \log d \quad \forall i \in I, \qquad \sum_{i \in I} x_{0,i} \leq X_0.$$

Then

$$0 \leq R_\gamma(x_\gamma) - L_C(x_0), \qquad R_\gamma(x_\gamma) - L_C(x_0) \leq \gamma\,\mathrm{KL}(x_0),$$

and also

$$R_\gamma(x_\gamma) - L_C(x_0) \leq \gamma \sum_{i \in I} \ell_i(x_0), \qquad R_\gamma(x_\gamma) - L_C(x_0) \leq \gamma X_0 \log d.$$

Proof. This is the finite theorem certified in Lean. Since x_γ is feasible and KL is nonnegative on $\mathcal{F}$, we have $R_\gamma(x_\gamma) \geq L_C(x_\gamma)$. Since x_0 minimizes L_C on $\mathcal{F}$, $L_C(x_\gamma) \geq L_C(x_0)$, proving the lower bound. Conversely, minimality of x_γ for the regularized objective gives

$$R_\gamma(x_\gamma) \leq R_\gamma(x_0) = L_C(x_0) + \gamma\,\mathrm{KL}(x_0),$$

which gives the bound by $\gamma\,\mathrm{KL}(x_0)$. The identity $\mathrm{KL}(x_0) = \sum_i \ell_i(x_0)$ is definitional and gives the third displayed bound. Finally, since I is nonempty we have $d \geq 1$ and hence $\log d \geq 0$, so

$$\sum_i \ell_i(x_0) \leq \sum_i x_{0,i} \log d = \left(\sum_i x_{0,i}\right) \log d \leq X_0 \log d,$$

and multiplying by $\gamma \geq 0$ yields the last bound. $\square$

For the paper's linear-programming specialization, take $\mathcal{F} = \{x \in \mathbb{R}^d_{++} : Ax = b\}$, $\mathrm{KL}(x) = \mathrm{KL}(x|z)$, let $x_0 = x_0^\star$ be an unregularized minimizer and let x_γ be a regularized minimizer. Then $L_C(x_0) = F_0^\star$ and $R_\gamma(x_\gamma) = F_\gamma^\star$, so

$$0 \leq F_\gamma^\star - F_0^\star \;\leq\; \gamma\,\mathrm{KL}(x_0^\star|z). \tag{11}$$

If moreover $z = (X_0^\star/d)\mathbb{1}$ and the coordinate KL terms satisfy $\ell_i(x_0^\star) \leq x_{0,i}^\star \log d$ with $\|x_0^\star\|_1 \leq X_0^\star$, Lemma B.1 also gives

$$0 \leq F_\gamma^\star - F_0^\star \;\leq\; \gamma\, X_0^\star \log d. \tag{12}$$

Proof of Theorem 3.2.

Proof. Split the total deviation into bias and optimisation pieces:

$$\left|F_0^\star - F_\gamma(u^{(k)})\right| \le \underbrace{|F_0^\star - F_\gamma^\star|}_{\text{bias}} + \underbrace{|F_\gamma^\star - F_\gamma(u^{(k)})|}_{\text{optimisation}}.$$

By the linear-programming specialization following Lemma B.1, the existence of $x_0^\star$ with $\|x_0^\star\|_1 \le X_0^\star$ implies $|F_0^\star - F_\gamma^\star| \le \gamma X_0^\star \log d$. With $\gamma = \varepsilon/(2X_0^\star \log d)$ this gives $F_0^\star - F_\gamma^\star \le \varepsilon/2$. The $O(1/k)$ dual rate of Theorem 3.1 yields $F_\gamma^\star - F_\gamma(u^{(k)}) \le \frac{16 X_\gamma U_\gamma^2 \|A\|_{1\to1}^2}{\gamma k}$. Our choice of $k \ge 32 X_\gamma U_\gamma^2 \|A\|_{1\to1}/(\gamma\varepsilon)$ ensures $F_\gamma^\star - F_\gamma(u^{(k)}) \le \varepsilon/2$. Summing the two contributions gives the claim. ☐

C Extension to general Bregman divergences

The analysis presented above for the KL divergence extends naturally to a broader class of Bregman divergences. We briefly summarize the main ingredients and results.

Bregman divergence. Let $\mathcal{X} \subset \mathbb{R}^d$ be a nonempty closed convex set with nonempty relative interior, and let $\phi : \mathrm{ri}(\mathcal{X}) \to \mathbb{R}$ be a Legendre-type convex function (proper, lower-semicontinuous, essentially smooth and strictly convex on $\mathrm{ri}(\mathcal{X})$). The Bregman divergence induced by ϕ is defined as

$$D_\phi(x|y) \coloneqq \phi(x) - \phi(y) - \langle \nabla\phi(y), x - y\rangle, \qquad x, y \in \mathrm{ri}(\mathcal{X}).$$

When $\mathcal{X} = \mathbb{R}_+^d$ and $\phi(x) = \sum_{i=1}^d x_i(\log x_i - 1)$, we recover the (non-normalized) KL divergence.

Bregman-regularized linear program. Fix a reference point $z \in \mathrm{ri}(\mathcal{X})$ and a temperature $\gamma > 0$. The Bregman-regularized problem reads

$$\min_{x \in \mathcal{C}_1 \cap \mathcal{C}_2} \langle C, x\rangle + \gamma\, D_\phi(x|z), \tag{$\widetilde{\mathcal{P}}_\gamma$}$$

which can be equivalently written as a Bregman projection onto $\mathcal{C}_1 \cap \mathcal{C}_2$ of the shifted reference $z_\gamma \coloneqq \nabla\phi^*\big(\nabla\phi(z) - \frac{C}{\gamma}\big)$.

Dual problem. Define the shifted dual reference $y_\gamma \coloneqq \nabla\phi(z) - \frac{C}{\gamma}$. The dual objective is

$$\widetilde{F}_\gamma(u) \;\coloneqq\; \langle b, u\rangle - \gamma\, \phi^*\Big(y_\gamma + \frac{A^\top u}{\gamma}\Big),$$

with the primal-dual relation $x(u) = \nabla\phi^*\big(y_\gamma + \frac{A^\top u}{\gamma}\big)$. The dual stationarity condition remains $A\, x(u^\star) = b$.

Generalized Pinsker condition. The key assumption replacing the Pinsker inequality is that there exists a primal confinement set $\mathcal{X}_\gamma \subset \mathrm{ri}(\mathcal{X})$, a norm $\|\cdot\|_*$ on $\mathbb{R}^d$, and a constant $\eta_\gamma > 0$ such that

$$\forall (x, y) \in \mathcal{X}_\gamma \times \mathcal{X}_\gamma, \qquad D_\phi(x|y) \ge \eta_\gamma\, \|x - y\|_*^2. \tag{$\widetilde{C}_\gamma$}$$

For the KL divergence with $\mathcal{X}_\gamma = \{x \in \mathbb{R}_{++}^d : \|x\|_1 \le X_\gamma\}$ and $\|\cdot\|_* = \|\cdot\|_1$, this holds with $\eta_\gamma = 1/(2X_\gamma)$.

Dual convergence result. With the generalized Pinsker condition ($\widetilde{C}_\gamma$) in place, the dual convergence result (Theorem 3.1) extends as follows. Let $\|A\|_{*\to\mathcal{V}^*}$ denote the operator norm from $(\mathbb{R}^d, \|\cdot\|_*)$ to the dual of the block-quotient norm. Assume the dual iterates satisfy $\sup_k \|u^{(k)}\|_V \le U_\gamma$ and that the primal iterates remain in $\mathcal{X}_\gamma$. Then

$$0 \le \Delta_k \le \frac{2\,\|A\|_{*\to\mathcal{V}^*}^2}{\gamma\,\eta_\gamma}\, U_\gamma^2\, \frac{1}{k}.$$

In the entropic case, $\eta_\gamma = 1/(2X_\gamma)$ and $\|A\|_{*\to\mathcal{V}^*} = \|A\|_{1\to1}$, recovering the bound of Theorem 3.1.

Bias bound. For the general Bregman case, the bias between the unregularized and regularized optima satisfies

$$0 \le F_\gamma^\star - F_0^\star \le \gamma\, D_\phi(x_0^\star | z),$$

where $x_0^\star$ is an optimal solution of the unregularized problem. Combined with the dual rate above, this yields iteration complexity bounds analogous to Theorem 3.2.

Remark C.1. *Beyond the entropic (KL) case, the Cressie–Read family of divergences provides another concrete example. For $\alpha \in (0,1)$, the generator $\varphi_\alpha(s) = \frac{s^\alpha - \alpha s + (\alpha-1)}{\alpha(\alpha-1)}$ yields a separable Bregman divergence on $\mathbb{R}^d_{++}$. Condition ($\widetilde{C}_\gamma$) then holds with $\eta_\gamma = \frac{1}{2d} X_\gamma^{\alpha-2}$ and $\|\cdot\|_* = \|\cdot\|_1$. Unlike the KL case ($\alpha = 1$), the curvature-based argument introduces a dimension factor $1/d$ in the Pinsker constant.*

C.1 Quantum optimal transport as a noncommutative Bregman projection

The general-Bregman framework naturally extends to entropic quantum optimal transport (QOT). This appendix presents a self-contained formulation aligned with the blueprint developed in the paper and clarifies what is already in place versus what remains open. In short, the variational and geometric setup is clear, but obtaining explicit, problem-dependent bounds for the key blueprint constants is still an open step. Matrix-valued and quantum variants of optimal transport have appeared in control and signal-processing formulations [Ning et al., 2015], in the mean-field/classical limits of quantum mechanics [Golse et al., 2016, Caglioti et al., 2020], and in quantum-information and learning settings [Chakrabarti et al., 2019]. Entropic regularization for matrix-valued/quantum OT was developed in [Peyré et al., 2019], which is the closest numerical precedent for the variational problem considered here.

Variational QOT problem. Let $\mathbb{H}_N$ denote Hermitian matrices, identify $N = nm$, and write $\mathrm{Tr}_2 : \mathbb{H}_{nm} \to \mathbb{H}_n$ and $\mathrm{Tr}_1 : \mathbb{H}_{nm} \to \mathbb{H}_m$ for the two partial traces. Given positive semidefinite marginals $A \in \mathbb{H}_{n,+}$ and $B \in \mathbb{H}_{m,+}$ with equal trace, and a Hermitian cost $C \in \mathbb{H}_{nm}$, the unregularized QOT problem is

$$\mathrm{W}_C(A,B) \coloneqq \min_{T \succeq 0} \{\mathrm{tr}(CT) \;:\; \mathrm{Tr}_2(T) = A,\ \mathrm{Tr}_1(T) = B\}.$$

Its dual maximizes $\mathrm{tr}(FA) + \mathrm{tr}(GB)$ under the Loewner constraint $F \otimes I_m + I_n \otimes G \preceq C$. The exact entropic regularization is obtained from the noncommutative entropy generator $\phi(T) = \mathrm{tr}(T \log T - T)$ and a reference $Z \succ 0$:

$$\min_{\mathrm{Tr}_2(T)=A,\ \mathrm{Tr}_1(T)=B,\ T\succ 0} \mathrm{tr}(CT) + \gamma D_\phi(T|Z), \qquad D_\phi(T|Z) = \mathrm{tr}\big(T(\log T - \log Z) - T + Z\big).$$

Equivalently, with $K_\gamma = \exp(\log Z - C/\gamma)$, this is the Bregman projection of K_γ onto the intersection of the two affine marginal constraints.

Dual and block maps. With dual variables $(F,G) \in \mathbb{H}_n \times \mathbb{H}_m$, the regularized dual objective is

$$\mathcal{F}_\gamma(F,G) = \mathrm{tr}(FA) + \mathrm{tr}(GB) - \gamma\, \mathrm{tr}\exp\left(\frac{F \otimes I_m + I_n \otimes G - C}{\gamma}\right)$$

up to the harmless reference-dependent shift when $Z \neq I$. The primal-dual relation is

$$T(F,G) = \exp\left(\frac{F \otimes I_m + I_n \otimes G - C}{\gamma}\right),$$

or, for general Z, the same formula with $-C/\gamma$ replaced by $\log Z - C/\gamma$. Alternating maximization defines exact block maps $\Phi_1(G) = \arg\max_F \mathcal{F}_\gamma(F,G)$ and $\Phi_2(F) = \arg\max_G \mathcal{F}_\gamma(F,G)$. These maps are implicit in general, because matrix logarithms and exponentials do not commute; this lack of a closed form is not itself fatal for the blueprint, which only requires exact block solves and quantitative control of the resulting sweep.

Quantum Pinsker geometry. The correct analogue of the classical Pinsker inequality is the quantum Pinsker inequality, historically attributed to Hiai–Ohya–Tsukada [Hiai et al., 1981]; it also follows from Uhlmann's earlier relative-entropy interpolation results [Uhlmann, 1977], and is stated in modern textbook form in [Watrous, 2018]. For trace-one $P, Q \succeq 0$,

$$D(P\|Q) \coloneqq \operatorname{tr}\bigl(P(\log P - \log Q)\bigr) \geq \frac{1}{2}\|P - Q\|_1^2, \qquad \|X\|_1 \coloneqq \operatorname{tr}|X|.$$

Thus the natural primal norm in the QOT blueprint is the trace/nuclear norm, and the natural dual norm is the operator norm. For the pair of dual variables one should quotient by the gauge $(F, G) \sim (F + cI_n, G - cI_m)$, leading to the spectral-variation seminorm

$$\|(F, G)\|_{\mathrm{var,op}} \coloneqq \inf_{c \in \mathbb{R}} \max\{\|F + cI_n\|_{\mathrm{op}}, \|G - cI_m\|_{\mathrm{op}}\}.$$

This is the direct noncommutative counterpart of the variation seminorm that appears for classical Sinkhorn and graph flow-Sinkhorn.

Open constants for the blueprint. The variational dictionary is now clear: QOT fits the general-Bregman proof with the nuclear norm on primal perturbations, the quotient operator norm on dual variables, and a Pinsker constant supplied by quantum Pinsker on trace slices. What remains open is to bound the problem-dependent constants needed to apply the generic convergence theorem. In particular, one needs a usable bound on the quantum analogue of H_γ, equivalently a lower spectral bound on the optimal T_γ, and a uniform quotient-operator bound on the dual iterates.

Why the classical monotonicity proof does not transfer. In the scalar Sinkhorn setting, monotonicity plus translation equivariance of the dual maps implies non-expansiveness in the variation seminorm. In QOT the natural order is the Loewner order, but this order is not a lattice order, and the exact block maps Φ_1, Φ_2 are not operator monotone in the sense needed to reproduce the topical-map argument. Preliminary numerical and differential evidence indicates that an individual block map need not be non-expansive in the spectral-variation seminorm, although it remains open whether the full sweep could satisfy a weaker orbit-confinement or non-expansiveness estimate. New strategies are therefore needed, for instance via derivative projection-strip estimates, scalar-shifted positive decompositions, or direct strong-concavity/cocoercivity bounds in the quotient operator geometry.

D Proofs for primal and dual iterate bounds

Bounding Primal and Dual Iterates. Our main result, Theorem 3.1, crucially relies on uniform primal and dual bounds, denoted respectively by $X_{\max}$ and $U_{\max}$. We first show in Subsection D that an $O(1/\gamma)$ bound on the primal variables automatically follows from a dual bound (in some cases, such as classical optimal transport, an $O(1)$ primal bound holds a priori, and this step can be skipped). Subsection D then presents a general strategy to establish dual boundedness, based on the assumption that the sweep map Ψ of the algorithm is non-expansive in the V-norm. This approach applies both to the classical optimal transport setting and to the optimal transport on graphs problem studied here.

Bounding Primal Iterates from Dual Iterates. In many cases, one directly has access to a primal bound X_γ (for instance, in classical OT, $X_\gamma = 1$). If this is not the case, the following proposition shows that it is always possible to derive a bound X_γ from U_γ, with the issue being that it blows as $X_\gamma \sim \|b\|_1 U_\gamma/\gamma$ when $\gamma \to 0$. The main workload is thus to bound the dual potential in the block quotient semi-norm, which needs to be done on a case-by-case basis and exploit the structure of the split (A_1, A_2).

Proof of Proposition 4.2.

Proof. Recall that $F_\gamma(u) = \langle b, u\rangle - \gamma\|x(u)\|_1$, hence $\|x^{(k)}\|_1 = \frac{\langle b, u^{(k)}\rangle - F_\gamma(u^{(k)})}{\gamma}$. Because $F_\gamma(u^{(k)})$ is nondecreasing along the ascent, we have $F_\gamma(u^{(k)}) \geq F_\gamma(u^{(0)}) = F_\gamma(0)$. Therefore,

$$\|x^{(k)}\|_1 \leq \frac{\langle b, u^{(k)}\rangle - F_\gamma(0)}{\gamma}. \tag{13}$$

We first bound $\langle b, u^{(k)} \rangle$ using the block-quotient control. Choose $h^{(k)} = (h_1^{(k)}, h_2^{(k)}) \in \ker(A^\top)$ such that $\|u^{(k)} + h^{(k)}\|_\infty \le \|u^{(k)}\|_V \le U_\gamma$. Since $b \in \mathrm{range}(A)$, one has $\langle b, h^{(k)} \rangle = 0$, hence

$$\langle b, u^{(k)} \rangle = \langle b, u^{(k)} + h^{(k)} \rangle \le \|b\|_1 \|u^{(k)} + h^{(k)}\|_\infty \le \|b\|_1 U_\gamma.$$

It remains to bound $-F_\gamma(0)$ explicitly. Since $u^{(0)} = 0$, $F_\gamma(0) = -\gamma \sum_{i=1}^d \exp(\frac{-C_i}{\gamma})$. Using $C_i \ge C_{\min}$ gives $\exp(-C_i/\gamma) \le \exp(-C_{\min}/\gamma)$, hence $F_\gamma(0) \ge -\gamma d\, e^{-C_{\min}/\gamma}$. Plugging the two estimates into (13) yields

$$\|x^{(k)}\|_1 \le \frac{\|b\|_1 U_\gamma + \gamma d\, e^{-C_{\min}/\gamma}}{\gamma} = \frac{\|b\|_1 U_\gamma}{\gamma} + d\, e^{-C_{\min}/\gamma},$$

which is the mass bound stated in Proposition 4.2. □

Bounding Dual Iterates using Non-expansiveness of Ψ. The most difficult part is to control the dual iterates. In this section, we provide a generic blueprint that leverages the non-expansiveness of Ψ. It requires a primal bound H_γ (measuring deviation from the reference z in dual coordinates) and a control of the conditioning of the splitting through a decomposition constant κ.

Definition D.1 (Primal bound H_γ in dual coordinates)**.** *We define $H_\gamma \in [0, +\infty]$ so that the minimizer x_γ of $(\mathcal{P}_\gamma)$ satisfies*

$$\|\log x_\gamma - \log z\|_\infty \le H_\gamma. \tag{14}$$

Definition D.2 (Decomposition constant κ)**.** *The decomposition constant is*

$$\kappa = \kappa(A_1, A_2) \coloneqq \sup_{y \in \mathrm{range}(A^\top),\, y \neq 0} \inf \left\{ \frac{\|w_1\|_\infty}{\|y\|_\infty} : \exists w_2 \text{ s.t. } A_1^\top w_1 + A_2^\top w_2 = y \right\} \in [0, +\infty]. \tag{15}$$

Proof of Proposition 4.1.

Proof. Let $u_\gamma = (u_{\gamma,1}, u_{\gamma,2})$ be any maximizer of (D_γ). Since u_γ is globally optimal, each block is optimal given the other, hence $u_{\gamma,1}$ is a fixed point of Ψ: $u_{\gamma,1} = \Psi(u_{\gamma,1})$. Triangular inequality with $\|\cdot\|_{V_1}$ gives

$$\|u_1^{(k)}\|_{V_1} \le \|u_1^{(k)} - u_{\gamma,1}\|_{V_1} + \|u_{\gamma,1}\|_{V_1}. \tag{16}$$

Using the iteration $u_1^{(k)} = \Psi^k(u_1^{(0)})$, the fixed-point property, and the non-expansiveness assumption, and triangular inequality

$$\|u_1^{(k)} - u_{\gamma,1}\|_{V_1} = \|\Psi^k(u_1^{(0)}) - \Psi^k(u_{\gamma,1})\|_{V_1} \le \|u_1^{(0)} - u_{\gamma,1}\|_{V_1} \le \|u_1^{(0)}\|_{V_1} + \|u_{\gamma,1}\|_{V_1}.$$

Combining with (16) gives

$$\|u_1^{(k)}\|_{V_1} \le \|u_1^{(0)}\|_{V_1} + 2\|u_{\gamma,1}\|_{V_1}. \tag{17}$$

Let $y_\gamma \coloneqq A^\top u_\gamma \in \mathrm{range}(A^\top)$. By the definition of κ in (15), there exist w_1, w_2 such that $A_1^\top w_1 + A_2^\top w_2 = y_\gamma$ and

$$\|w_1\|_\infty \le \kappa\, \|y_\gamma\|_\infty. \tag{18}$$

Set $h_1 \coloneqq w_1 - u_{\gamma,1}$ and $h_2 \coloneqq w_2 - u_{\gamma,2}$. Then $(h_1, h_2) \in \ker(A^\top)$, hence by definition of $\|\cdot\|_{V_1}$,

$$\|u_{\gamma,1}\|_{V_1} \le \|u_{\gamma,1} + h_1\|_\infty = \|w_1\|_\infty \le \kappa\, \|y_\gamma\|_\infty. \tag{19}$$

It remains to bound $\|y_\gamma\|_\infty$. At a dual maximizer u_γ, stationarity gives $Ax_\gamma = b$ with $x_\gamma = x(u_\gamma)$. Moreover, by definition of $x(u)$, componentwise, $A^\top u_\gamma = C + \gamma(\log x_\gamma - \log z)$. Using (14), we have $\|\log x_\gamma - \log z\|_\infty \le H_\gamma$, hence

$$\|A^\top u_\gamma\|_\infty \le \|C\|_\infty + \gamma \|\log x_\gamma - \log z\|_\infty \le \|C\|_\infty + \gamma H_\gamma.$$

Inserting this into (19) yields

$$\|u_{\gamma,1}\|_{V_1} \le \kappa \left(\|C\|_\infty + \gamma H_\gamma\right).$$

Combining with (17) gives the bound stated in Proposition 4.1. □

E Balanced optimal transport specialization

This section does not present new contributions, it revisits the classical entropic approximation of discrete optimal transport (OT). This "warm-up" illustrates how the constants in Theorem 3.1 specialise to familiar quantities and recovers the standard $O(C_{\max}/k)$ Sinkhorn rate. The next subsection extends the methodology to the graph–based W_1 distance (Section 5).

Problem set-up. For OT, the primal vector is a transport plan $x = P \in \mathbb{R}_{++}^{m_1 \times m_2}$, so that the dimension is $d = m_1 m_2$. The two marginals $b_1 \in \mathbb{R}_{++}^{m_1}$ and $b_2 \in \mathbb{R}_{++}^{m_2}$ are probability vectors, $\sum_i b_{1i} = \sum_j b_{2j} = 1$. A coupling $P \in \mathbb{R}_{++}^{m_1 \times m_2}$ must satisfy

$$A_1(P) = b_1, \qquad A_2(P) = b_2, \quad \text{where} \quad A_1(P) \coloneqq P\mathbf{1}_{m_2},\ A_2(P) \coloneqq P^\top \mathbf{1}_{m_1}.$$

With a cost matrix $C \in \mathbb{R}_{+}^{m_1 \times m_2}$ the unregularised OT instance reads

$$\mathrm{W}_C(b_1, b_2) \coloneqq \min_{P \geq 0, A(P) = b} \langle C, P \rangle. \tag{20}$$

Entropic dual and Sinkhorn updates. The primal-dual relation between solution u of $(\mathcal{D}_\gamma)$ and $P(u)$ of $(\mathcal{P}_\gamma)$ with reference vector $z = b_1 \otimes b_2 = (b_{1,i} b_{2,j})_{i,j}$ (note that we use a separable reference measure for simplicity) reads

$$P(u)_{i,j} = b_{1,i} b_{2,j} e^{\frac{u_{1,i} + u_{2,j} - C_{i,j}}{\gamma}}.$$

The Sinkhorn update, written over the dual variable, corresponds to the two block updates where Ψ_1, Ψ_2 are soft-c-transforms

$$\Psi_1(u_2)_i \coloneqq -\gamma \log \sum_j e^{\frac{-C_{i,j} + u_{2,j}}{\gamma}} b_{2,j}, \quad \Psi_2(u_1)_j \coloneqq -\gamma \log \sum_i e^{\frac{-C_{i,j} + u_{1,i}}{\gamma}} b_{1,i}. \tag{21}$$

The block-quotient semi-norm is the so-called variation distance defined in (26):

$$\| \cdot \|_{V_1} = \| \cdot \|_{V_2} = \| \cdot \|_{\mathrm{Var}}.$$

The following propositions give the value for the primal bound H_γ of Definition 4.1 and the decomposition constant κ defined in (4), which are the constants involved in the convergence rates. Recall that for OT, the reference measure is $z = b_1 \otimes b_2 = (b_{1,i} b_{2,j})_{i,j}$.

Proposition E.1 (Sinkhorn log-ratio H_γ certificate)**.** *Let I, J be finite nonempty index sets, let $b : J \to \mathbb{R}$ be a positive probability vector, let $r : I \times J \to \mathbb{R}$ be a log-ratio field, let $u : I \to \mathbb{R}$ and $v : J \to \mathbb{R}$ be nonnegative scaling vectors, and let $C : I \times J \to \mathbb{R}$ be a cost field. Let $\min_b, C_{\max}, \gamma \in \mathbb{R}$ satisfy $0 < \min_b$ and $\gamma > 0$. Assume that $0 \leq C_{i,j} \leq C_{\max}$ for every (i, j), that $\min_b \leq b_j$ for every $j \in J$, and that the normalized row-scaling identities*

$$\sum_{j \in J} b_j e^{r_{i,j}} = 1 \qquad \textit{hold for every } i \in I.$$

Assume also the finite Sinkhorn scaling certificates

$$v_j \sum_{i \in I} e^{-C_{i,j}/\gamma} u_i = b_j \qquad \forall j \in J,$$

and

$$e^{r_{i,j}} = \frac{e^{-C_{i,j}/\gamma}}{\left(\sum_{\ell \in J} e^{-C_{i,\ell}/\gamma} v_\ell\right) \left(\sum_{k \in I} e^{-C_{k,j}/\gamma} u_k\right)} \qquad \forall (i, j).$$

Then, with

$$H_\gamma = |\log(\min_b)| + \frac{2C_{\max}}{\gamma},$$

one has $H_\gamma \geq 0$ and $|r_{i,j}| \leq H_\gamma$ for every (i, j). For classical Sinkhorn, take $r_{i,j} = \log(P_\gamma)_{i,j} - \log z_{i,j}$, $b = b_2$, $\min_b = \min(\min_i (b_1)_i, \min_j (b_2)_j)$, and $C_{\max} = \|C\|_\infty$.

Proof. The displayed implication is the finite theorem certified in Lean. In the paper-facing statement, the probability-vector normalization of b, the nonnegativity of the scaling vectors u, v, and the finite cost envelope $0 \le C_{i,j} \le C_{\max}$ are represented as typed finite-data certificates. First, the probability-vector assumptions imply $\min_b \le 1$: choose $j_0 \in J$, then $\min_b \le b_{j_0} \le \sum_j b_j = 1$, where the second inequality uses positivity of the other coordinates. For fixed i, j, every term in $\sum_\ell b_\ell e^{r_{i,\ell}}$ is nonnegative and the sum is equal to one, hence $b_j e^{r_{i,j}} \le 1$. Since $0 < \min_b \le b_j$, this gives

$$e^{r_{i,j}} \le \frac{1}{b_j} \le \frac{1}{\min_b}, \qquad \text{so} \qquad r_{i,j} \le \log(1/\min_b) = -\log(\min_b).$$

Because $0 < \min_b \le 1$, we have $-\log(\min_b) = |\log(\min_b)|$. Thus the row-scaling equation supplies the upper bound $r_{i,j} \le |\log(\min_b)| \le H_\gamma$. Set $K_{i,j} = e^{-C_{i,j}/\gamma}$ and

$$D_{i,j} := \left(\sum_{\ell \in J} K_{i,\ell} v_\ell\right)\left(\sum_{k \in I} K_{k,j} u_k\right).$$

Since $\gamma > 0$ and $0 \le C_{i,j} \le C_{\max}$, Lean first derives the Gibbs-kernel bounds

$$K_{i,j} \le 1, \qquad \exp(-C_{\max}/\gamma) \le K_{i,j} \qquad \forall (i,j).$$

Lean formalizes the finite-sum estimate

$$D_{i,j} \le \Big(\sum_{\ell \in J} v_\ell\Big)\Big(\sum_{k \in I} u_k\Big) \le \exp(C_{\max}/\gamma).$$

Indeed, the first inequality follows from $K_{i,j} \le 1$ and nonnegativity of u, v. The right marginal scaling equations imply the total scaled mass

$$\sum_{k,\ell} u_k K_{k,\ell} v_\ell = \sum_\ell v_\ell (K^T u)_\ell = \sum_\ell b_\ell = 1.$$

For the second inequality, the lower kernel bound gives

$$1 = \sum_{k,\ell} u_k K_{k,\ell} v_\ell \ge \exp(-C_{\max}/\gamma)\Big(\sum_k u_k\Big)\Big(\sum_\ell v_\ell\Big).$$

The equality $e^{r_{i,j}} = K_{i,j}/D_{i,j}$, together with $e^{r_{i,j}} > 0$ and $K_{i,j} > 0$, also implies $D_{i,j} > 0$. Hence

$$e^{-2C_{\max}/\gamma} D_{i,j} \le e^{-2C_{\max}/\gamma} e^{C_{\max}/\gamma} = e^{-C_{\max}/\gamma} \le K_{i,j},$$

and, since $D_{i,j} > 0$, one has

$$e^{-2C_{\max}/\gamma} \le \frac{K_{i,j}}{D_{i,j}} = e^{r_{i,j}}.$$

By monotonicity of the exponential, $-2C_{\max}/\gamma \le r_{i,j}$, and therefore $-H_\gamma \le -2C_{\max}/\gamma \le r_{i,j}$. Hence $|r_{i,j}| \le H_\gamma$ for every coordinate. The nonnegativity of H_γ follows from $|\log(\min_b)| \ge 0$, $C_{\max} \ge 0$, and $\gamma > 0$.

We now recall why the Sinkhorn specialization supplies these certificates. Let P_γ be the unique minimizer of the entropic OT problem and recall that $z_{i,j} = (b_1)_i (b_2)_j$. We need to bound $|\log(P_\gamma)_{i,j} - \log z_{i,j}|$ uniformly over all (i,j).

Upper bound. From the optimality conditions, $P_\gamma = z \odot \exp((f \oplus g - C)/\gamma)$ where f, g are the Sinkhorn potentials satisfying the marginal constraints. The row marginal gives

$$\sum_j (b_2)_j \exp\left(\frac{f_i + g_j - C_{i,j}}{\gamma}\right) = 1.$$

Since all terms are non-negative and sum to 1, each term is at most 1: $(b_2)_j \exp\left(\frac{f_i+g_j-C_{i,j}}{\gamma}\right) \le 1$, hence

$$\log(P_\gamma)_{i,j} - \log z_{i,j} = \frac{f_i + g_j - C_{i,j}}{\gamma} \le -\log((b_2)_j) \le -\log(\min_b) = |\log(\min_b)|.$$

The same bound follows from the column marginal using $(b_1)_i$.

Lower bound. The optimal coupling reads $P_\gamma = \mathrm{diag}(u)K\,\mathrm{diag}(v)$ for scaling vectors u, v and $K_{i,j} := e^{-C_{i,j}/\gamma}$. Since $0 \le C_{i,j} \le \|C\|_\infty$, this kernel satisfies $K_{i,j} \le 1$ and $K_{i,j} \ge K_{\min} := e^{-\|C\|_\infty/\gamma}$. From $P_\gamma = \mathrm{diag}(u)K\,\mathrm{diag}(v)$ and the marginal constraints, $(b_1)_i = u_i(Kv)_i$, $(b_2)_j = v_j(K^\top u)_j$, hence

$$(P_\gamma)_{i,j} = u_i K_{i,j} v_j = \frac{(b_1)_i (b_2)_j\, K_{i,j}}{(Kv)_i\,(K^\top u)_j}. \tag{22}$$

Since $K \le \mathbb{1}\mathbb{1}^\top$ entrywise, $(Kv)_i \le \sum_\ell v_\ell$ and $(K^\top u)_j \le \sum_k u_k$. Also, the total mass constraint gives

$$1 = \sum_{a,b} (P_\gamma)_{a,b} = u^\top K v \ge K_{\min}\Big(\sum_k u_k\Big)\Big(\sum_\ell v_\ell\Big),$$

so $\big(\sum_k u_k\big)\big(\sum_\ell v_\ell\big) \le 1/K_{\min}$ and therefore $(Kv)_i\,(K^\top u)_j \le \frac{1}{K_{\min}}$. Moreover $K_{\min} = e^{-\|C\|_\infty/\gamma}$ and, for $D_{i,j} = (Kv)_i(K^\top u)_j$, equation (22) gives $e^{r_{i,j}} = K_{i,j}/D_{i,j}$ with $0 < D_{i,j} \le e^{\|C\|_\infty/\gamma}$ and $K_{i,j} \ge e^{-\|C\|_\infty/\gamma}$. These are exactly the scaling facts derived from the cost bounds in the certified statement. Equivalently, plugging the denominator bound into (22) yields the familiar scalar lower estimate $(P_\gamma)_{i,j} \ge (b_1)_i(b_2)_j\, K_{i,j}\, K_{\min} \ge z_{i,j}\, K_{\min}^2$, hence

$$\log(P_\gamma)_{i,j} - \log z_{i,j} \ge 2\log K_{\min} = -\frac{2\|C\|_\infty}{\gamma}.$$

Thus the classical OT specialization satisfies the proposition with $C_{\max} = \|C\|_\infty$. □

Proposition E.2 (Constructive κ certificate for classical OT)**.** *Let I_1, I_2 be finite nonempty index sets. Let $\alpha : I_1 \to \mathbb{R}$, $\beta : I_2 \to \mathbb{R}$, and $Y : I_1 \times I_2 \to \mathbb{R}$ satisfy*

$$Y_{i,j} = \alpha_i + \beta_j \qquad \forall i \in I_1,\ j \in I_2.$$

Fix any $j_0 \in I_2$. Then there exist $w_1 : I_1 \to \mathbb{R}$ and $w_2 : I_2 \to \mathbb{R}$ such that

$$Y_{i,j} = w_{1,i} + w_{2,j} \qquad \forall i, j, \qquad \|w_1\|_\infty \le \|Y\|_\infty.$$

In particular, this constructive decomposition certificate implies that for classical OT one can take $\kappa = 1$ in (4).

Proof. Choose any representation $Y_{ij} = \alpha_i + \beta_j$. For any scalar $c \in \mathbb{R}$, define $\alpha^{(c)} := \alpha - c\,\mathbb{1}_{I_1}, \beta^{(c)} := \beta + c\,\mathbb{1}_{I_2}$, so that $\alpha_i^{(c)} + \beta_j^{(c)} = \alpha_i + \beta_j = Y_{ij}$ for all (i, j). We now show that one can choose c so that $\|\alpha^{(c)}\|_\infty \le \|Y\|_\infty$. Indeed,

$$\min_{c\in\mathbb{R}} \|\alpha - c\mathbb{1}_{I_1}\|_\infty = \frac{1}{2}\left(\max_i \alpha_i - \min_i \alpha_i\right).$$

Using the fixed column index $j_0 \in I_2$, for all i, i',

$$\alpha_i - \alpha_{i'} = (\alpha_i + \beta_{j_0}) - (\alpha_{i'} + \beta_{j_0}) = Y_{ij_0} - Y_{i'j_0}.$$

Hence $|\alpha_i - \alpha_{i'}| \le 2\|Y\|_\infty$, which implies $\max_i \alpha_i - \min_i \alpha_i \le 2\|Y\|_\infty$. Therefore, $\min_{c\in\mathbb{R}} \|\alpha - c\mathbb{1}_{I_1}\|_\infty \le \|Y\|_\infty$. Choose c attaining (or arbitrarily approximating) this minimum and set $w_1 := \alpha^{(c)}$, $w_2 := \beta^{(c)}$. Then $w_{1,i} + w_{2,j} = Y_{ij}$ and $\|w_1\|_\infty \le \|Y\|_\infty$. For a nonzero Y this gives

$$\inf\left\{\frac{\|w_1\|_\infty}{\|Y\|_\infty} :\ \exists w_2,\ A_1^\top w_1 + A_2^\top w_2 = Y\right\} \le 1.$$

Taking the supremum over $Y \in \mathrm{range}(A^\top) \setminus \{0\}$ gives $\kappa \le 1$. □

Corollary E.1 (OT zero-start orbit constants)**.** *Let I, J be finite nonempty index sets and let*

$$\Psi_1 : \mathbb{R}^J \to \mathbb{R}^I, \qquad \Psi_2 : \mathbb{R}^I \to \mathbb{R}^J$$

be the two Sinkhorn block maps. Assume the two block maps are monotone and satisfy the signed block translation-equivariance laws of Proposition G.3; consequently their full sweep is $\Psi = \Psi_1 \circ \Psi_2 :$

$\mathbb{R}^I \to \mathbb{R}^I$. Let $\alpha^\star \in \mathbb{R}^I$ be a fixed point of this full sweep. Assume that there exist $\beta^\star \in \mathbb{R}^J$, a reference index $j_0 \in J$, and $Y : I \times J \to \mathbb{R}$ such that

$$\alpha_i^\star + \beta_j^\star = Y_{i,j} \qquad \forall i \in I,\ j \in J.$$

Let $\gamma, \min_b, C_{\max} \in \mathbb{R}$ satisfy $\gamma > 0$, $\min_b > 0$, and $C_{\max} \geq 0$. Assume the separable fixed-point budget

$$\|Y\|_\infty \leq C_{\max} + \gamma \left(|\log(\min_b)| + \frac{2C_{\max}}{\gamma} \right).$$

Then, for the zero potential initialization, there exists a certificate $X_\gamma = 1$ and, for all $k \geq 0$,

$$|\Psi^k(0)|_V \leq U_\gamma \coloneqq 6\, C_{\max} + 2\gamma |\log(\min_b)|.$$

For classical OT, set $\min_b \coloneqq \min(\min_i (b_1)_i, \min_j (b_2)_j)$ and $C_{\max} = \|C\|_\infty$. With the zero potential initialization, the assumptions above are supplied by Propositions E.2 and E.1, so one may take $X_\gamma = 1$ and $U_\gamma = 6\,\|C\|_\infty + 2\gamma|\log(\min_b)|$.

Proof. The displayed finite implication is the Lean-certified statement. In Lean the two block maps, their monotonicity laws, and their signed translation-equivariance laws are grouped in a structured block-sweep certificate. The fixed point, the auxiliary vector $\beta^\star$, the reference index j_0, the separable decomposition $\alpha_i^\star + \beta_j^\star = Y_{i,j}$, and the displayed budget are grouped in a separable fixed-point certificate. The scalar side conditions $\gamma > 0$, $\min_b > 0$, and $C_{\max} \geq 0$ are grouped in a third bookkeeping certificate. These records only organize the hypotheses displayed above; they do not add hidden assumptions. The block monotonicity and signed translation-equivariance assumptions imply topicality of the full sweep by Propositions G.2 and G.3. Hence Proposition G.1 gives non-expansiveness for the variation seminorm. The separable certificate $\alpha_i^\star + \beta_j^\star = Y_{i,j}$, together with Proposition E.2, gives

$$|\alpha^\star|_V \leq \|Y\|_\infty \leq C_{\max} + \gamma \left(|\log(\min_b)| + \frac{2C_{\max}}{\gamma} \right).$$

Applying Proposition 4.1 with $H_\gamma = |\log(\min_b)| + 2C_{\max}/\gamma$ and $\kappa = 1$ gives

$$|\Psi^k(0)|_V \leq 2 \left(C_{\max} + \gamma \left(|\log(\min_b)| + \frac{2C_{\max}}{\gamma} \right) \right) = 6C_{\max} + 2\gamma |\log(\min_b)|,$$

because the zero potential has variation seminorm zero. The certificate $X_\gamma = 1$ records the normalized total mass of classical OT couplings. For the classical OT specialization, Proposition E.1 supplies the displayed H_γ bound with $C_{\max} = \|C\|_\infty$. □

This bound U_γ derived using the non-expansiveness of Ψ is not sharp. As shown in Chizat et al. [2020], a more direct argument exploiting the closed-form expression (21) for Ψ, together with the Lipschitz dependence of the dual variables on the cost matrix C, yields the tighter estimate $U_\gamma = \frac{\|C\|_\infty}{2}$.

Using Corollary E.1, Theorem 3.2 shows that Sinkhorn achieves ε-additive accuracy on the optimal transport cost using $O(\frac{n^2}{\varepsilon^2}\, \|C\|_\infty^2 \log n)$ arithmetic operations. This matches exactly the complexity bounds established in Altschuler et al. [2017], Chakrabarty and Khanna [2021], Dvurechensky et al. [2018], Chizat et al. [2020].

F Proofs for the graph W_1 Sinkhorn-flow algorithm

This appendix contains the proofs associated with Section 5. We keep the notation of the main text: E is the directed edge set, $p = |E|$, $x = (f, g) \in \mathbb{F}^2$, $\mathcal{C}_1$ is the divergence constraint, and $\mathcal{C}_2$ is the equality constraint $f = g$.

The linear operators are $A_1(f, g) = f\mathbb{1}_n - g^\top \mathbb{1}_n$ and $A_2(f, g) = f - g$, and for dual variables $(v, U) \in \mathbb{R}^n \times \mathbb{F}$ the adjoint action is

$$A^\top(v, U) = \Big((v_i + U_{i,j})_{(i,j) \in E},\ (-v_j - U_{i,j})_{(i,j) \in E} \Big) \in \mathbb{F} \times \mathbb{F}. \tag{23}$$

Proof of Proposition 5.1. For the projection onto $\mathcal{C}_1$, the KKT conditions for $\min_{f,g} \mathrm{KL}((f,g)|(h,h))$ subject to $-f\mathbb{1}_n + g^\top \mathbb{1}_n = b_1 - b_2$ give a multiplier $\lambda \in \mathbb{R}^n$ such that $\log(f_{i,j}/h_{i,j}) + \lambda_i = 0$ and $\log(g_{i,j}/h_{i,j}) - \lambda_j = 0$. Setting $s = e^{-\lambda}$ gives $f = \mathrm{diag}(s)h$ and $g = h\,\mathrm{diag}(s)^{-1}$. Inserting these expressions in the divergence constraint yields $(s \odot s) \odot (h\mathbb{1}_n) + s \odot (b_1 - b_2) - h^\top \mathbb{1}_n = 0$, whose positive root is the function ϕ of Proposition 5.1. The projection onto $\mathcal{C}_2$ is a diagonal KL projection, hence the geometric mean.

Proof of Proposition 5.2. Combining the two closed-form projections just proved with the parametrization $f^{(k)} = \mathrm{diag}(s^{(k)}) z^C \mathrm{diag}(1/s^{(k)})$ gives the scaling update of Proposition 5.2. Passing to $v = 2\gamma \log s$ and writing the sums in log-sum-exp form gives the stable formula (9).

Proposition F.1 (Closed forms for $\|\cdot\|_{V_1}$ and $\|\cdot\|_{V_2}$)**.** *After the connected-graph kernel identification in the proof below, the quotient directions on the vertex block and edge block are the constant directions. Equivalently, for any finite nonempty vertex index set V and edge index set E, for every $v \in \mathbb{R}^V$ and $U \in \mathbb{R}^E$,*

$$\|v\|_{V_1} = \inf_{c\in\mathbb{R}} \|v + c\mathbb{1}_V\|_\infty = \|v\|_{\mathrm{Var}}, \qquad \|U\|_{V_2} = \inf_{c\in\mathbb{R}} \|U + c\mathbb{1}_E\|_\infty = \|U\|_{\mathrm{Var}},$$

where $\|\cdot\|_{\mathrm{Var}}$ is the variation semi-norm defined in (26)*.*

Proposition F.2 (Topical graph-flow sweep is non-expansive)**.** *Let V be a finite nonempty vertex set and let $\Psi_1, \Psi_2 : \mathbb{R}^V \to \mathbb{R}^V$ be topical maps, i.e. monotone and translation–equivariant in the sense of Proposition G.1. Then the full sweep map $\Psi = \Psi_1 \circ \Psi_2$ is non-expansive for the variation quotient norm on the vertex block:*

$$\|\Psi(v) - \Psi(w)\|_{\mathrm{Var}} \le \|v - w\|_{\mathrm{Var}} \qquad \textit{for all } v, w \in \mathbb{R}^V.$$

In the graph-flow split, the required topicality certificates are obtained by taking the diagonal signature $\Sigma = \mathrm{diag}(+I_E, -I_E)$ on the two flow blocks and the translation parameter $\tau = +1$.

Proof. The first sentence is the theorem certified in Lean: each topical map is non-expansive for $\|\cdot\|_{\mathrm{Var}}$ by Proposition G.1, and the composition of two maps non-expansive for the same seminorm is again non-expansive.

It remains to recall why the graph-flow split supplies the topicality certificates used above. With the sign convention detailed below, multiplying the g block by -1 turns each column of the two block moment maps into a nonnegative incidence contribution: the $v_i + U_{i,j}$ and $-v_j - U_{i,j}$ terms become monotone in the signed coordinates. Thus $A_s\Sigma$ is entrywise nonnegative for the two block updates. Moreover the two signed incidence contributions cancel after adding a constant to the vertex potential and the same constant to the edge multiplier, giving the paired-balance identity of Proposition G.3 with $\tau = +1$. Proposition G.2 gives monotonicity of the block maps and Proposition G.3 gives translation equivariance of the sweep. These are precisely the graph-flow topicality certificates invoked in the proposition. □

Proposition F.3 (Closed form and non-expansiveness of Ψ_2)**.** *Let V be a finite nonempty vertex set, let E be a finite nonempty directed edge set, and let $s, t : E \to V$ be its source and target endpoint maps. Define the closed-form second-block map $\Psi_2 : \mathbb{R}^V \to \mathbb{R}^E$ by*

$$\Psi_2(v)_e = \frac{1}{2}\big(v_{t(e)} - v_{s(e)}\big).$$

Then, for every $v \in \mathbb{R}^V$,

$$\Psi_2(v)_e = \frac{1}{2}\big(v_{t(e)} - v_{s(e)}\big) \quad \textit{for all } e \in E, \qquad \|\Psi_2(v)\|_{\mathrm{Var}} \le \|v\|_{\mathrm{Var}}. \tag{24}$$

Through Proposition F.1, this is the quotient-norm bound $\|\Psi_2(v)\|_{V_2} \le \|v\|_{V_1}$ used in the graph-W_1 analysis.

Proof of Proposition F.1. The kernel relation $(\delta v, \delta U) \in \ker(A^\top)$ is equivalent to $\delta v_i + \delta U_{i,j} = 0$ and $-\delta v_j - \delta U_{i,j} = 0$ for every $(i,j) \in E$. Since the graph is connected, this implies $\delta v = c\mathbb{1}_n$ and $\delta U = -c\mathbb{1}_E$. Therefore Definition 3.1 gives $\|v\|_{V_1} = \inf_c \|v + c\mathbb{1}_n\|_\infty = \|v\|_{\mathrm{Var}}$ and $\|U\|_{V_2} = \inf_c \|U - c\mathbb{1}_E\|_\infty = \|U\|_{\mathrm{Var}}$.

Proof of Proposition F.3. Maximizing $F_\gamma(v, U)$ with respect to U_e at fixed v gives the displayed closed-form map. For this map,

$$|\Psi_2(v)_e| = \frac{1}{2}|v_{t(e)} - v_{s(e)}| \le \frac{1}{2}(\max_i v_i - \min_i v_i) = \|v\|_{\mathrm{Var}}.$$

The definition of $\|\cdot\|_{\mathrm{Var}}$ as an infimum over constant shifts therefore gives $\|\Psi_2(v)\|_{\mathrm{Var}} \le \|v\|_{\mathrm{Var}}$. Proposition F.1 translates this variation-seminorm estimate into the quotient-norm estimate.

Proposition F.4 (Graph-W_1 log-ratio H_γ certificate)**.** *Let E be a finite nonempty edge set, let $e \mapsto \bar{e}$ be an opposite-orientation map on E, let $f, z : E \to \mathbb{R}_{++}$, and let $W : E \to \mathbb{R}$. Let $X_\star, W_{\max}, \gamma, L_z \in \mathbb{R}$ satisfy $\gamma > 0$ and $W_{\max} \ge 0$. Assume that, for every $e \in E$,*

$$f_e \le X_\star, \qquad |\log z_e| \le L_z, \qquad W_e + W_{\bar{e}} \le 2W_{\max},$$

and assume the opposite-orientation log identity

$$\log f_e + \log f_{\bar{e}} = \log z_e + \log z_{\bar{e}} - \frac{W_e + W_{\bar{e}}}{\gamma}. \tag{25}$$

Then, with

$$H_\gamma = \log X_\star + \frac{2W_{\max}}{\gamma} + 3L_z,$$

one has $H_\gamma \ge 0$ and

$$|\log f_e - \log z_e| \le H_\gamma \qquad \forall e \in E.$$

In the graph-flow application, $f = f_\gamma$, z is the positive reference flow, $L_z = \|\log z\|_\infty$, and

$$X_\star = X_\gamma^\star := \frac{\langle W, \bar{f}\rangle + \gamma \mathrm{KL}(\bar{f}|z)}{W_{\min}},$$

where $\bar{f} \ge 0$ is any feasible comparison flow and $0 < W_{\min} := \min_{e \in E} W_e$.

Proof. This is the finite theorem certified in Lean. Set $r_e = \log f_e - \log z_e$. Since $f : E \to \mathbb{R}_{++}$ and $f_e \le X_\star$, monotonicity of the logarithm gives $\log f_e \le \log X_\star$; together with $|\log z_e| \le L_z$, this yields

$$r_e \le \log X_\star + L_z.$$

For the lower bound, rearrange the opposite-orientation identity (25) as

$$r_e = \log z_{\bar{e}} - \log f_{\bar{e}} - \frac{W_e + W_{\bar{e}}}{\gamma}.$$

Using $|\log z_{\bar{e}}| \le L_z$, $0 < f_{\bar{e}} \le X_\star$, and $W_e + W_{\bar{e}} \le 2W_{\max}$ gives

$$-\Big(\log X_\star + \frac{2W_{\max}}{\gamma} + L_z\Big) \le r_e.$$

Combining the upper and lower estimates gives

$$-H_\gamma \le r_e \le H_\gamma, \qquad H_\gamma = \log X_\star + \frac{2W_{\max}}{\gamma} + 3L_z,$$

because $L_z \ge 0$, which follows from nonemptiness of E and $0 \le |\log z_e| \le L_z$ for any edge e. Hence $|\log f_e - \log z_e| \le H_\gamma$ for every edge. Since E is nonempty, applying this bound to one edge and using $|\log f_e - \log z_e| \ge 0$ also gives $H_\gamma \ge 0$. In the graph-flow specialization, Lemma F.1 supplies the mass envelope $f_e \le X_\gamma^\star$, while the optimality condition of the two-orientation graph block gives (25). □

Lemma F.1 (Finite primal ℓ^1 bound under positive costs)**.** *Let I be a finite nonempty coordinate set. Let $\mathcal{A}$ be any additional constraint predicate on $\mathbb{R}^I$, define the feasible set*

$$\mathcal{F} := \{x \in \mathbb{R}^I : \ x_i \ge 0 \ \forall i \in I, \ \mathcal{A}(x)\},$$

and let $\ell_i : \mathbb{R}^I \to \mathbb{R}$, $i \in I$, be finite KL coordinate terms. Set

$$\mathrm{KL}(x) := \sum_{i \in I} \ell_i(x).$$

Fix $C, x^\star_\gamma, \bar{x} \in \mathbb{R}^I$ and $\gamma > 0$. Assume the costs are strictly positive, $C_i > 0$ for all $i \in I$, and define

$$C_{\min} := \min_{i \in I} C_i.$$

Assume that the finite KL terms are nonnegative on the feasible set:

$$\ell_i(x) \geq 0 \qquad \forall x \in \mathcal{F},\ \forall i \in I.$$

Assume $x^\star_\gamma \in \mathcal{F}$ minimizes

$$x \longmapsto \sum_{i \in I} C_i x_i + \gamma \mathrm{KL}(x)$$

over $\mathcal{F}$, and assume $\bar{x} \in \mathcal{F}$. Then

$$\sum_{i \in I} (x^\star_\gamma)_i \leq \frac{\sum_{i \in I} C_i \bar{x}_i + \gamma \mathrm{KL}(\bar{x})}{C_{\min}}.$$

In the graph-W_1 application, $\mathcal{F}$ is the affine nonnegative feasible set, $\mathrm{KL} = \mathrm{KL}(\cdot|z)$ is the finite sum of scalar KL terms, and $\bar{x}$ is any feasible comparison flow.

Proof. Since I is finite nonempty and $C_i > 0$ for every coordinate, the finite minimum satisfies $C_{\min} > 0$ and $C_{\min} \leq C_i$ for every $i \in I$; this finite-minimum step is part of the Lean proof. Since $x^\star_\gamma \in \mathcal{F}$, its coordinatewise nonnegativity follows directly from the definition of $\mathcal{F}$. By coordinatewise positivity,

$$C_{\min} \sum_i (x^\star_\gamma)_i \leq \sum_i C_i (x^\star_\gamma)_i.$$

Since $x^\star_\gamma \in \mathcal{F}$, the coordinatewise KL-term nonnegativity gives

$$\mathrm{KL}(x^\star_\gamma) = \sum_i \ell_i(x^\star_\gamma) \geq 0.$$

Together with $\gamma \geq 0$, this implies

$$\sum_i C_i (x^\star_\gamma)_i \leq \sum_i C_i (x^\star_\gamma)_i + \gamma \mathrm{KL}(x^\star_\gamma).$$

The minimizing property of $x^\star_\gamma$ over $\mathcal{F}$, applied to $\bar{x} \in \mathcal{F}$, bounds the last quantity by $\sum_i C_i \bar{x}_i + \gamma \mathrm{KL}(\bar{x})$. Dividing by $C_{\min} > 0$ gives the claim. $\square$

Proposition F.5 (Graph-W_1 rooted path κ certificate)**.** *Let I be a finite nonempty vertex index set. Let $\kappa, B \in \mathbb{R}$ with $0 \leq B \leq 1$, let $D \in \mathbb{N}$, let $y^f, y^g : I \times I \to \mathbb{R}$, and let $(P_i)_{i \in I}$ be a rooted path family, where each P_i is a finite list of oriented pairs in $I \times I$. Assume*

$$|y^f_p| \leq B, \qquad |y^g_p| \leq B \quad \forall p \in I \times I, \qquad |P_i| \leq D \quad \forall i \in I,$$

and that one rooted path controls the quotient constant:

$$\exists i \in I, \qquad \kappa \leq \left| \sum_{p \in P_i} \left(y^f_p + y^g_p \right) \right|.$$

Then

$$\kappa \leq 2D.$$

Proof. This is the rooted-path certificate formalized in Lean. Along any path element,

$$|y^f_p + y^g_p| \leq |y^f_p| + |y^g_p| \leq 2B \leq 2.$$

Thus a path with at most D elements has summed edge-gradient magnitude at most $2D$, and the assumed quotient-control inequality gives $\kappa \leq 2D$.

For $y = A^\top(v, U)$ in the graph-W_1 specialization, write $y^{(f)}_{i,j} = v_i + U_{i,j}$ and $y^{(g)}_{i,j} = -v_j - U_{i,j}$. The edge field $g_{i,j} = y^{(f)}_{i,j} + y^{(g)}_{i,j} = v_i - v_j$ satisfies $\|g\|_\infty \leq 2\|y\|_\infty$. Fixing a root and integrating this gradient along shortest paths supplies the rooted path family with $D = \mathrm{diameter}(E)$; this is the graph-theoretic specialization. $\square$

Corollary F.1 (Graph-W_1 zero-start iterate constants)**.** *Let I be a finite nonempty vertex index set and let J be an auxiliary block index set. Let*

$$\Psi_1 : \mathbb{R}^J \to \mathbb{R}^I, \qquad \Psi_2 : \mathbb{R}^I \to \mathbb{R}^J$$

be the two graph-W_1 block maps. Assume that Ψ_1 and Ψ_2 are monotone and satisfy the signed block translation-equivariance laws of Proposition G.3 for some paired sign τ. Set $\Psi = \Psi_1 \circ \Psi_2$ and let $v^\star$ be a fixed point of this full sweep. Let $\gamma > 0$, $b_{\mathrm{mass}} \geq 0$, $0 \leq B \leq 1$, and assume $\ell_{\max} + \gamma H_\gamma \geq 0$. Let $y^f, y^g : I \times I \to \mathbb{R}$ satisfy

$$|y^f_q| \leq B, \qquad |y^g_q| \leq B \qquad \forall q \in I \times I.$$

Suppose also that

$$\|v^\star\|_{\mathrm{Var}} \leq \kappa(\ell_{\max} + \gamma H_\gamma).$$

If a finite list of edge-gradient increments $(s_a)_a$ satisfies

$$\#\{a\} \leq D, \qquad s_a = y^f_{p_a} + y^g_{p_a} \text{ for some } p_a \in I \times I, \qquad \kappa \leq \left| \sum_a s_a \right|,$$

and if the primal mass proxy satisfies, for all $k \geq 0$,

$$X(k) \leq \frac{b_{\mathrm{mass}} \, \|\Psi^k(0)\|_{\mathrm{Var}}}{\gamma} + p\, e^{-W_{\min}/\gamma},$$

then there exist constants

$$U_\gamma = 4D(\ell_{\max} + \gamma H_\gamma), \qquad X_\gamma = \frac{b_{\mathrm{mass}} U_\gamma}{\gamma} + pe^{-W_{\min}/\gamma}$$

such that, for all $k \geq 0$,

$$\|\Psi^k(0)\|_{\mathrm{Var}} \leq U_\gamma, \qquad X(k) \leq X_\gamma.$$

In the graph-W_1 specialization, $D = \mathrm{diameter}(E)$, $\ell_{\max} = W_{\max}$, and $b_{\mathrm{mass}} = \|b\|_1$.

Proof. The Lean statement for this corollary uses named certificate records for the two block maps and their monotonicity/translation laws, the fixed point and its H_γ-κ budget, the bounded edge fields y^f, y^g, the finite two-step path list, and the pointwise mass proxy. These records are only organizational: their fields are exactly the hypotheses displayed above. Topicality gives non-expansiveness in the variation seminorm, hence Proposition 4.1, applied from the zero initialization, gives

$$\|\Psi^k(0)\|_{\mathrm{Var}} \leq 2\|v^\star\|_{\mathrm{Var}}.$$

The fixed-point budget and the path certificate yield

$$\|\Psi^k(0)\|_{\mathrm{Var}} \leq 2\,\kappa(\ell_{\max} + \gamma H_\gamma) \leq 4D(\ell_{\max} + \gamma H_\gamma) = U_\gamma.$$

Substituting this bound into the displayed mass proxy gives the stated X_γ. In the graph-W_1 application, Proposition F.2 supplies topicality, Proposition F.4 supplies the H_γ budget, and Proposition 4.2 supplies the mass proxy. $\square$

Proof of Theorem 5.1. Choose a spanning tree and route the signed measure $q = b_1 - b_2$ along this tree: for each edge of the tree, the flow is the total signed mass of one component after cutting that edge. This produces a feasible flow $\bar{f}$ with total transported mass at most $\mathrm{diameter}(E)\|q\|_1/2$ and cost at most $W_{\max}\mathrm{diameter}(E)\|q\|_1/2$. Therefore the unregularized optimum has an explicit feasible mass bound $X_0^\star = O(\mathrm{diameter}(E))$ for probability inputs. Lemma F.1, Proposition F.4, and Proposition F.5 give explicit H_γ, $U_\gamma = O(\mathrm{diameter}(E))$, and $X_\gamma = O(\mathrm{diameter}(E)/\gamma + pe^{-W_{\min}/\gamma})$. With $\gamma \asymp \varepsilon$ and $p = o(1/\log(1/\varepsilon))$, the exponential term is lower order. Theorem 3.2 then requires $O(X_\gamma U_\gamma^2/\varepsilon^2) = O(\mathrm{diameter}(E)^3/\varepsilon^4)$ iterations up to logarithmic factors, and each iteration costs $O(p)$ sparse edge operations.

G Non-expansiveness in Variation Semi-norm

Topical maps and non-expansiveness. We first recall a classical result of so-called "topical maps" in the nonlinear Perron–Frobenius/max–plus theory Candrall and Tartar [1980]. This ensures the non-expansiveness of a map for the variation seminorm, which is ℓ^∞ norm quotiented by translation

$$\|v\|_{\mathrm{Var}} \coloneqq \inf_{c\in\mathbb{R}} \|v + c\,\mathbb{1}\|_\infty = \frac{1}{2}\mathrm{osc}(v) \quad \text{where} \quad \mathrm{osc}(v) = \max(v) - \min(v), \tag{26}$$

where $\mathbb{1}$ denotes the all-ones vector of the appropriate dimension. The remaining part of this section shows that this result can be applied to $T = \Psi$, the sweep dual mapping. This result is pivotal to show dual boundedness as exposed in Section D.

Proposition G.1 (Monotone, translation–equivariant maps are non–expansive in the V–seminorm)**.** *Let $T : \mathbb{R}^n \to \mathbb{R}^n$ satisfy:*

1. Monotonicity: $x \le y$ *coordinatewise* $\Rightarrow T(x) \le T(y)$ *coordinatewise.*
2. Translation–equivariance: $T(x + c\mathbb{1}) = T(x) + c\mathbb{1}$ *for all* $x \in \mathbb{R}^n$ *and* $c \in \mathbb{R}$.

Then T is non–expansive for $\|\cdot\|_{\mathrm{Var}}$:

$$\|T(x) - T(y)\|_{\mathrm{Var}} \le \|x - y\|_{\mathrm{Var}} \qquad \textit{for all } x, y \in \mathbb{R}^n.$$

Proof. Fix $x, y \in \mathbb{R}^n$ and set $d \coloneqq y - x \in \mathbb{R}^n$. Let $a \coloneqq \min_{1\le i\le n} d_i, b \coloneqq \max_{1\le i\le n} d_i$, so that $a \le d_i \le b$ for every i, i.e. coordinatewise, $x + a\mathbb{1} \le y \le x + b\mathbb{1}$ By monotonicity of T and translation equivariance this implies

$$T(x + a\mathbb{1}) \le T(y) \le T(x + b\mathbb{1}) \quad \Rightarrow \quad T(x) + a\mathbb{1} \le T(y) \le T(x) + b\mathbb{1},$$

hence for each coordinate i, $a \le \big(T(y) - T(x)\big)_i \le b$. Therefore

$$\max_i \big(T(y) - T(x)\big)_i \le b, \qquad \min_i \big(T(y) - T(x)\big)_i \ge a,$$

and taking the oscillation gives

$$\mathrm{osc}\big(T(y) - T(x)\big) \le b - a = \max_i d_i - \min_i d_i = \mathrm{osc}(d) = \mathrm{osc}(y - x).$$

Dividing by 2 yields the desired non–expansiveness. $\square$

Monotonicity. We assume there exists a diagonal signature matrix $\Sigma = \mathrm{diag}(\sigma_1, \dots, \sigma_d)$ with $\sigma_i \in \{\pm 1\}$ such that

$$B_s \coloneqq A_s\Sigma \quad \text{is entrywise nonnegative, for } s \in \{1, 2\}. \tag{27}$$

Equivalently, $A_s = B_s\Sigma$ with $B_s \ge 0$ componentwise.

Using Σ and (27), define a partial order $\preceq_\Sigma$ on $\mathbb{R}^{m_2}$ by

$$u_2 \preceq_\Sigma v_2 \iff \begin{cases} (B_2^\top u_2)_i \le (B_2^\top v_2)_i & \text{for all } i \text{ with } \sigma_i = +1, \\ (B_2^\top u_2)_i \ge (B_2^\top v_2)_i & \text{for all } i \text{ with } \sigma_i = -1. \end{cases} \tag{28}$$

Proposition G.2 (Composition of anti-monotone block updates)**.** *Let R_2 be a relation on the second block and let $\Psi = \Psi_1 \circ \Psi_2$. Assume the two block anti-monotonicity laws*

$$R_2(u_2, v_2) \quad \Longrightarrow \quad \Psi_1(v_2) \le \Psi_1(u_2), \tag{29}$$

$$u_1 \le v_1 \quad \Longrightarrow \quad R_2(\Psi_2(v_1), \Psi_2(u_1)). \tag{30}$$

Then the two displayed laws hold, and the full sweep is monotone:

$$u_1 \le v_1 \quad \Longrightarrow \quad \Psi(u_1) = \Psi_1(\Psi_2(u_1)) \le \Psi_1(\Psi_2(v_1)) = \Psi(v_1).$$

In the signed KL setting of (27)*, this proposition is applied with $R_2 = \preceq_\Sigma$.*

Proof. The abstract composition step is immediate. If $u_1 \le v_1$, then (30) gives $R_2(\Psi_2(v_1), \Psi_2(u_1))$, and (29) gives $\Psi_1(\Psi_2(u_1)) \le \Psi_1(\Psi_2(v_1))$. This is the order-theoretic theorem certified in Lean. We now recall why, in the signed KL setting, the relation $R_2 = \preceq_\Sigma$ satisfies the two anti-monotonicity laws.

For $u \in \mathbb{R}^m$, the primal-dual relation reads $x_i(u) = z_i^C \exp((A^\top u)_i)$. Introducing the moment maps M_s, the first-order optimality conditions read

$$M_s(\Psi_s(u)) = b_s \quad \text{where} \quad M_s(u) := A_s x(u) \in \mathbb{R}^{m_s}, \tag{31}$$

Using (27), write $A_s = B_s\Sigma$ with $B_s \ge 0$ componentwise. Then for each i,

$$x_i(u) = z_i^C \exp\Big(\frac{\sigma_i}{\gamma}\big((B_1^\top u_1)_i + (B_2^\top u_2)_i\big)\Big), \tag{32}$$

(i) Anti-monotonicity of Ψ_1. Assume $u_2 \preceq_\Sigma v_2$. Let $u_1 := \Psi_1(u_2), v_1 := \Psi_1(v_2)$. Then $M_1(u_1, u_2) = b_1$ and $M_1(v_1, v_2) = b_1$ by (31). By Lemma G.1, for any fixed w,

$$M_1(w, u_2) \le M_1(w, v_2) \qquad \text{(componentwise)}. \tag{33}$$

In particular, $M_1(v_1, u_2) \le M_1(v_1, v_2) = b_1$. Suppose for contradiction that $v_1 \not\le u_1$, i.e. there exists an index j with $(v_1)_j > (u_1)_j$. Since $u_1 \mapsto M_1(u_1, u_2)$ is componentwise nondecreasing by Lemma G.1, we obtain

$$M_1(v_1, u_2) \ge M_1(u_1, u_2) = b_1.$$

Together with $M_1(v_1, u_2) \le b_1$, this yields a contradiction. Hence $v_1 \le u_1$, i.e. $\Psi_1(v_2) \le \Psi_1(u_2)$.

(ii) Anti-monotonicity of Ψ_2 *in the signed order.* Assume $u_1 \le v_1$. Let

$$u_2 := \Psi_2(u_1), \qquad v_2 := \Psi_2(v_1).$$

Then $M_2(u_1, u_2) = b_2$ and $M_2(v_1, v_2) = b_2$ by (31). By Lemma G.1, for any fixed w, $M_2(u_1, w) \le M_2(v_1, w)$ (componentwise). In particular, $M_2(u_1, v_2) \le M_2(v_1, v_2) = b_2$. Now suppose for contradiction that $v_2 \not\preceq_\Sigma u_2$. Since $u_2 \mapsto M_2(u_1, u_2)$ is nondecreasing in the signed order by Lemma G.1, the failure of $v_2 \preceq_\Sigma u_2$ implies $M_2(u_1, v_2) \not\le M_2(u_1, u_2) = b_2$ (and in fact $M_2(u_1, v_2) \ge b_2$ componentwise). This contradicts $M_2(u_1, v_2) \le b_2$. Hence $v_2 \preceq_\Sigma u_2$, i.e. $\Psi_2(v_1) \preceq_\Sigma \Psi_2(u_1)$.

(iii) Monotonicity of Ψ. Assume $u_1 \le v_1$. By (ii), $\Psi_2(v_1) \preceq_\Sigma \Psi_2(u_1)$. Apply (i), equivalently (29) with $R_2 = \preceq_\Sigma$, with $u_2 = \Psi_2(v_1)$ and $v_2 = \Psi_2(u_1)$ to obtain $\Psi_1(\Psi_2(u_1)) \le \Psi_1(\Psi_2(v_1))$, i.e. $\Psi(u_1) \le \Psi(v_1)$. ◻

Lemma G.1 (Monotonicity of finite nonnegative moment maps)**.** *Let* S, I, J *be finite index sets. Let* $A : S \times I \to \mathbb{R}$ *and* $B : I \times J \to \mathbb{R}$ *satisfy* $A_{r,i} \ge 0$ *and* $B_{i,j} \ge 0$ *for all* r, i, j*. If* $x, y \in \mathbb{R}^S$ *satisfy* $x_r \le y_r$ *for every* $r \in S$*, then the two-layer moment map*

$$\mathcal{M}(x)_j = \sum_{i \in I} B_{i,j} \left(\sum_{r \in S} A_{r,i} x_r \right)$$

is componentwise monotone:

$$\mathcal{M}(x)_j \le \mathcal{M}(y)_j \qquad \text{for every } j \in J.$$

In the signed KL setting of (27)*, applying this finite nonnegative-layer statement to the signed source coordinates gives the monotonicity of* M_s *in the first block and the monotonicity of* M_s *with respect to* $\preceq_\Sigma$ *in the second block.*

Proof. Finite nonnegative-layer step. For each $i \in I$, nonnegativity of $A_{r,i}$ gives

$$\sum_{r \in S} A_{r,i} x_r \le \sum_{r \in S} A_{r,i} y_r.$$

Multiplying by $B_{i,j} \ge 0$ and summing over i gives $\mathcal{M}(x)_j \le \mathcal{M}(y)_j$ for every $j \in J$.

Signed KL application. We now recover the moment-map monotonicity used in the proof of Proposition G.2. Fix u_2. If $u_1 \le v_1$ componentwise, then $B_1^\top u_1 \le B_1^\top v_1$ componentwise because $B_1 \ge 0$. Hence for each i:

- if $\sigma_i = +1$, then the exponent in (32) increases, so $x_i(u_1, u_2) \le x_i(v_1, u_2)$ and $\sigma_i x_i(\cdot) B_{s,i}$ increases (since $B_{s,i} \ge 0$);
- if $\sigma_i = -1$, then the exponent decreases, so $x_i(u_1, u_2) \ge x_i(v_1, u_2)$, and multiplying by $\sigma_i = -1$ reverses the inequality: $\sigma_i x_i(u_1, u_2) \le \sigma_i x_i(v_1, u_2)$, hence again $\sigma_i x_i(\cdot) B_{s,i}$ increases componentwise.

This is exactly the finite nonnegative-layer argument above, with the signed atomwise contributions as the ordered intermediate layer. Summing over i yields $M_s(u_1, u_2) \le M_s(v_1, u_2)$ componentwise for $s \in \{1, 2\}$.

(2) monotonicity in u_2 for the signed order. Fix u_1. If $u_2 \preceq_\Sigma v_2$, then by definition (28),

$$(B_2^\top u_2)_i \le (B_2^\top v_2)_i \text{ when } \sigma_i = +1, \qquad (B_2^\top u_2)_i \ge (B_2^\top v_2)_i \text{ when } \sigma_i = -1.$$

Equivalently,

$$\sigma_i (B_2^\top u_2)_i \le \sigma_i (B_2^\top v_2)_i \qquad \text{for all } i.$$

Therefore the exponent in (32) increases for every i, which implies $x_i(u_1, u_2) \le x_i(u_1, v_2)$. Now consider the contributions $\sigma_i x_i(\cdot)\, B_{s,i}$:

- if $\sigma_i = +1$, increasing x_i increases the nonnegative vector $x_i B_{s,i}$;
- if $\sigma_i = -1$, increasing x_i decreases the nonpositive vector $-x_i B_{s,i}$, i.e. increases it componentwise.

Again applying the finite nonnegative-layer argument to the resulting signed intermediate layer and summing over i yields $M_s(u_1, u_2) \le M_s(u_1, v_2)$ componentwise for $s \in \{1, 2\}$, which is precisely nondecreasingness with respect to $\preceq_\Sigma$. □

Translation equivariance. Fix a sign parameter $\tau \in \{+1, -1\}$. We say that the pair (A_1, A_2) satisfies the *signed paired–balance condition* (with sign τ) if

$$A_1^\top \mathbb{1}_{m_1} + \tau\, A_2^\top \mathbb{1}_{m_2} = 0 \in \mathbb{R}^d. \tag{34}$$

One has $\tau = -1$ for classical OT and $\tau = +1$ for the lifted W_1 flow formulation on graphs.

Proposition G.3 (Translation equivariance from paired–balance block laws)**.** *Fix $\tau \in \{+1, -1\}$ and let $\mathcal{P}$ be a paired–balance certificate. Assume that $\mathcal{P}$ holds and that it implies the two signed block-translation laws*

$$\Psi_2(u_1 + c\,\mathbb{1}_{m_1}) = \Psi_2(u_1) + \tau\, c\,\mathbb{1}_{m_2}, \qquad \Psi_1(u_2 + c\,\mathbb{1}_{m_2}) = \Psi_1(u_2) + \tau\, c\,\mathbb{1}_{m_1} \tag{35}$$

for all admissible u_1, u_2, c. Then the two displayed block laws hold, and the full sweep $\Psi = \Psi_1 \circ \Psi_2 : \mathbb{R}^{m_1} \to \mathbb{R}^{m_1}$ is translation–equivariant:

$$\Psi(u_1 + c\,\mathbb{1}_{m_1}) = \Psi(u_1) + c\,\mathbb{1}_{m_1} \qquad \textit{for all } u_1, c. \tag{36}$$

In the concrete KL setting, the certificate $\mathcal{P}$ is the signed paired–balance identity (34)*.*

Proof. The abstract implication is the theorem certified in Lean. Since $\mathcal{P}$ holds, the two certified derivations give the two block laws in (35). Applying the second law to $u_1 + c\mathbb{1}_{m_1}$ gives a shift by τc in the second block. Applying the first law with shift τc then gives a first-block shift by $\tau(\tau c) = c$. Therefore $\Psi(u_1 + c\mathbb{1}_{m_1}) = \Psi(u_1) + c\mathbb{1}_{m_1}$.

It remains only to recall why the concrete paired–balance identity (34) provides the block-law derivations used above. Condition (34) implies

$$F_\gamma(u_1 + c\mathbb{1}_{m_1}, w) = F_\gamma(u_1, w - \tau c\mathbb{1}_{m_2}) + \langle b_1, c\mathbb{1}_{m_1}\rangle + \langle b_2, \tau c\mathbb{1}_{m_2}\rangle, \tag{37}$$

where the last two terms are constants independent of w. Hence, maximizing over w,

$$\arg\max_w F_\gamma(u_1 + c\mathbb{1}_{m_1}, w) = \arg\max_w F_\gamma(u_1, w - \tau c\mathbb{1}_{m_2}).$$

which is the desired result for Ψ_2, the proof for Ψ_1 being similar. □

H Lean Formalization Guide

The Lean development is intended as an audit trail for the mathematical structure of the paper. Each theorem, proposition, lemma, and corollary appearing in the manuscript is assigned a stable Lean alias, and the aliases are checked against compiled Lean theorem constants. The current paper-facing map covers all 27 manuscript statements, including the main convergence chain, the regularized approximation theorem, the OT and graph-W_1 instantiations, and the auxiliary ingredients used in the appendices. These include KL bias bounds, quotient-seminorm estimates, non-expansiveness interfaces, the per-step ascent estimate of Lemma A.1, the quotient residual estimate of Lemma A.2, and the non-normalised Pinsker reduction of Lemma A.3. Several auxiliary components are deliberately stated in reusable finite-dimensional forms, so that they can be inspected independently of the particular flow-Sinkhorn application.

How to navigate the Lean code. The current umbrella entry point is `lean/FlowSinkhorn/KLProjection.lean`, with project root `lean/FlowSinkhorn.lean`. For paper-oriented reading, the facade `lean/FlowSinkhorn/Paper.lean` imports section and appendix modules following the manuscript structure. The canonical synchronization layer is `lean/FlowSinkhorn/KLProjection/StatementMap.lean`; it is re-exported by `lean/FlowSinkhorn/Paper/StatementMap.lean`. In this map, each paper-facing name, such as `thm_3_1`, `lem_A_1`, or `prop_F_5`, is an alias for one canonical Lean theorem constant, and each alias carries an implementation-file comment indicating where the proof is defined. The map itself is intentionally proof-free: it is a stable index from manuscript statements to proof-producing Lean modules.

The implementation modules are organized by mathematical role rather than by LaTeX order. The main groups are duality and primal–dual identities, dual convergence and rate estimates, primal/dual uniform bounds, finite-dimensional variation geometry, and the OT and graph-W_1 application layers. This organization keeps reusable proof infrastructure separate from the paper-facing alias layer while still allowing a reader to start from a statement label and jump directly to the corresponding proof file.

Certification scope. The certified KL-projection backend currently contains 30,435 non-comment, non-blank lines of Lean code, with 1,633 theorem/lemma declarations and 63 direct definition/structure/class/abbrev/inductive declarations under the repository audit counter. The paper-facing map is checked by scripts that verify alias completeness, compiled endpoint existence, statement numbering, and implementation-file locations. The build is green for `cd lean && lake build FlowSinkhorn.KLProjection.StatementMap`, and the proof-producing KL-projection development contains no `sorry`, `admit`, or local `axiom` declarations.

Comparator challenge and solution. The repository also contains a Comparator-ready statement-checking layer. The trusted challenge file `lean/FlowSinkhorn/Comparator/Challenge.lean` imports only `Mathlib` and proof-free `FlowSinkhorn.Comparator.Vocabulary.*` modules, and it contains exactly the 27 reviewed statement-only declarations. The untrusted solution file `lean/FlowSinkhorn/Comparator/Solution.lean` imports only `FlowSinkhorn.KLProjection.StatementMap` and proves the same 27 theorem statements by the corresponding paper-facing aliases. The generated manifest, review dossier, and challenge lock record the paper labels, theorem names, implementation-file locations, statement hashes, and audit verdicts. Local checks verify that the challenge and solution statements match, that the challenge does not expose implementation theorem endpoints, and that the independent paper-to-challenge audit has 27 faithful entries and no qualified entries. A local fake-`landrun` Comparator smoke test accepts the solution; the remaining external step for a fully hardened Comparator certificate is to run the frozen configuration on Linux with real `landrun`.

A few examples illustrate what is certified. The formalized Pinsker appendix constructs the finite sign selector, relates it to the ℓ^1 distance, builds the two-point Bernoulli measure used for the sign test, invokes mathlib's measure-theoretic Hoeffding lemma, normalizes the common-mass variational inequality, performs the scalar quadratic optimization, and scales the result back to the non-normalised form used in Appendix A.3. The per-step ascent formalization combines this Pinsker layer with KL-gain certificates for the two block updates and then composes the two half-steps into the full sweep inequality of Lemma A.1. The quotient-residual formalization proves the finite Holder step, the shifted-representative and gauge-orthogonality manipulations, and the conversion from quotient-seminorm radii to the gap estimate used in Lemma A.2.

The Lean formalization was carried out after the authors had completed and checked the LaTeX proofs. It was assisted by ChatGPT (GPT-5.4) for proof generation and proof checking. The formalization did not lead to significant changes in the authors' mathematical arguments, but it did detect and help correct arithmetic constant mistakes in intermediate bounds.

I Notation

Notation	**Meaning**
d	Ambient dimension of the primal variable $x \in \mathbb{R}^d_+$.
$A = (A_1; A_2)$	Constraint matrix split into two blocks.
$b = (b_1; b_2)$	Right-hand side split compatibly with A_1, A_2.
$C \in \mathbb{R}^d$	Linear cost vector in the unregularized linear program.
$z \in \mathbb{R}^d_{++}$	Positive reference measure/vector for the KL penalty.
$\gamma > 0$	Entropic regularization parameter.
$\mathcal{C}_1, \mathcal{C}_2$	Affine constraint blocks $\{A_1 x = b_1\}$ and $\{A_2 x = b_2\}$.
$\mathrm{KL}(x\|z)$	Non-normalized Kullback–Leibler divergence from x to z.
z^C	Gibbs reference vector, $z_i^C = z_i \exp(-C_i/\gamma)$.
$F_\gamma(u)$	Dual objective associated with the entropically regularized problem.
$u = (u_1, u_2)$	Dual variable split according to the two constraint blocks.
$x(u)$	Primal variable recovered from a dual variable by the primal–dual relation.
Ψ_1, Ψ_2	Exact dual block maximization maps.
$\Psi = \Psi_1 \circ \Psi_2$	Full dual sweep map on the first dual block.
$\\|\cdot\\|_{V_1}, \\|\cdot\\|_{V_2}$	Block quotient seminorms induced by $\ker(A^\top)$.
$\\|\cdot\\|_V$	Maximum of the two block quotient seminorms.
Δ_k	Dual suboptimality gap $F_\gamma^\star - F_\gamma(u^{(k)})$.
X_γ	Uniform ℓ^1 bound on primal iterates.
U_γ	Uniform block-quotient bound on dual iterates.
H_γ	Uniform bound on $\\|\log x_\gamma - \log z\\|_\infty$ at the regularized optimum.
$\kappa(A_1, A_2)$	Decomposition constant converting control of $A^\top u$ into quotient control of dual blocks.
V, E	Vertex and edge sets of the graph used for graph W_1.
n, p	Number of graph vertices and directed sparse edge entries, respectively.
W	Edge-length/cost matrix on the graph.
$\mathbb{F}$	Sparse nonnegative flow cone supported on graph edges.
f, g	Duplicated graph-flow variables used in the flow-Sinkhorn splitting.
v, U	Vertex and edge dual variables in the graph-flow formulation.
$\mathrm{diameter}(E)$	Maximum shortest-path distance between graph vertices.
Σ, τ	Signature matrix and scalar balance parameter used for signed monotonicity and translation equivariance.
D_ϕ	General Bregman divergence generated by a convex function ϕ.
η_γ	Generalized Pinsker constant in the Bregman extension.